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New Revelations of the Americas

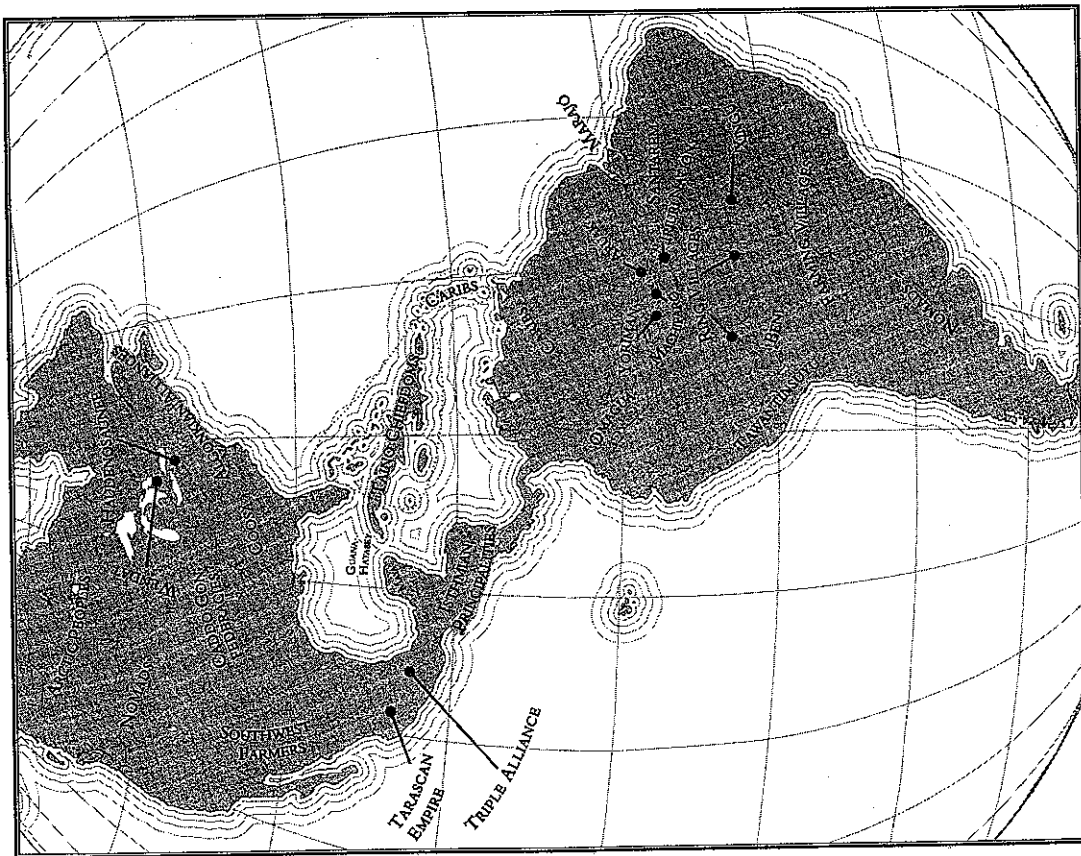


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New York

2006



NATIVE AMERICA, 1491 A.D.

PREFACE

The seeds of this book date back, at least in part, to 1983, when I wrote an article for *Science* about a NASA program that was monitoring atmospheric ozone levels. In the course of learning about the program, I flew with a research team in a NASA plane equipped to sample and analyze the atmosphere at thirty thousand feet. At one point the group landed in Mérida, in Mexico's Yucatán Peninsula. For some reason the scientists had the next day off, and we all took a decrepit Volkswagen van to the Maya ruins of Chichén Itzá. I knew nothing about Mesoamerican culture—I may not even have been familiar with the term "Mesoamerica," which encompasses the area from central Mexico to Panama, including all of Guatemala and Belize, and parts of El Salvador, Honduras, Costa Rica, and Nicaragua, the homeland of the Maya, the Olmec, and a host of other indigenous groups. Moments after we clambered out of the van I was utterly enthralled.

On my own—sometimes for vacation, sometimes on assignment—I returned to Yucatán five or six times, three times with my friend Peter Menzel, a photojournalist. For a German magazine, Peter and I made a twelve-hour drive down a terrible dirt road (thigh-deep potholes, blockades of fallen timber) to the then-unexcavated Maya metropolis of Calakmul. Accompanying us was Juan de la Cruz Briceño, Maya himself, caretaker of another, smaller ruin. Juan had spent twenty years as a *chiclero*, trekking the forest for weeks on end in search of chicle trees, which have a gooey sap that Indians have dried and chewed for millennia and that in the late nineteenth century became the base of the chewing-gum industry. Around a night fire he told us about the ancient, vine-shrouded cities he had stumbled across in his rambles, and his amazement when scientists informed him that his ancestors had built them. That night we slept in hammocks amid tall, headstone-like carvings that had not been read for more than a thousand years.

My interest in the peoples who walked the Americas before Columbus only snapped into anything resembling focus in the fall of 1992. By chance one Sunday afternoon I came across a display in a college library of the special Columbian quincentenary issue of the *Annals of the Association of American Geographers*. Curious, I picked up the journal, sank into an armchair, and

began to read an article by William Denevan, a geographer at the University of Wisconsin. The article opened with the question, "What was the New World like at the time of Columbus?" Yes, I thought, what *was* it like? Who lived here and what could have passed through their minds when European sails first appeared on the horizon? I finished Denevan's article and went on to others and didn't stop reading until the librarian flicked the lights to signify closing time.

I didn't know it then, but Denevan and a host of fellow researchers had spent their careers trying to answer these questions. The picture they have emerged with is quite different from what most Americans and Europeans think, and still little known outside specialist circles.

A year or two after I read Denevan's article, I attended a panel discussion at the annual meeting of the American Association for the Advancement of Science. Called something like "New Perspectives on the Amazon," the session featured William Balée of Tulane University. Balée's talk was about "anthropogenic" forests—forests *created* by Indians centuries or millennia in the past—a concept I'd never heard of before. He also mentioned something that Denevan had discussed: many researchers now believe their predecessors underestimated the number of people in the Americas when Columbus arrived. Indians were more numerous than previously thought, Balée said—much more numerous. Gee, someone ought to put all this stuff together, I thought. It would make a fascinating book.

I kept waiting for that book to appear. The wait grew more frustrating when my son entered school and was taught the same things I had been taught, beliefs I knew had long been sharply questioned. Since nobody else appeared to be writing the book, I finally decided to try it myself. Besides, I was curious to learn more. The book you are holding is the result.

Some things this book is not. It is not a systematic, chronological account of the Western Hemisphere's cultural and social development before 1492. Such a book, its scope vast in space and time, could not be written—by the time the author approached the end, new findings would have been made and the beginning would be outdated. Among those who assured me of this were the very researchers who have spent much of the last few decades wrestling with the staggering diversity of pre-Columbian societies.

Nor is this book a full intellectual history of the recent changes in perspective among the anthropologists, archaeologists, ecologists, geographers, and historians who study the first Americans. That, too, would be impossible, for the ramifications of the new ideas are still rippling outward in too many directions for any writer to contain them in one single work.

Instead, this book explores what I believe to be the three main foci of the

new findings: Indian demography (Part I), Indian origins (Part II), and Indian ecology (Part III). Because so many different societies illustrate these points in such different ways, I could not possibly be comprehensive. Instead, I chose my examples from cultures that are among the best documented, or have drawn the most recent attention, or just seemed the most intriguing.

Throughout this book, as the reader already will have noticed, I use the term "Indian" to refer to the first inhabitants of the Americas. No question about it, Indian is a confusing and historically inappropriate name. Probably the most accurate descriptor for the original inhabitants of the Americas is Americans. Actually using it, though, would be risking worse confusion. In this book I try to refer to people by the names they call themselves. The overwhelming majority of the indigenous peoples whom I have met in both North and South America describe themselves as Indians. (For more about nomenclature, see Appendix A, "Loaded Words.")

In the mid-1980s I traveled to the village of Hazelton, on the upper Skeena River in the middle of British Columbia. Many of its inhabitants belong to the Gitksan (or Gitksan) nation. At the time of my visit, the Gitksan had just lodged a lawsuit with the governments of both British Columbia and Canada. They wanted the province and the nation to recognize that the Gitksan had lived there a long time, had never left, had never agreed to give their land away, and had thus retained legal title to about eleven thousand square miles of the province. They were very willing to negotiate, they said, but they were not willing to not be negotiated with.

Flying in, I could see why the Gitksan were attached to the area. The plane swept past the snowy, magnificent walls of the Rocher de Boule Mountains and into the confluence of two forested river valleys. Mist steamed off the land. People were fishing in the rivers for steelhead and salmon even though they were 165 miles from the coast.

The Gitksan band of the Gitksan has its headquarters in Hazelton, but most members live in a reserve just outside town. I drove to the reserve, where Neil Sterritt, head of the Gitksan council, explained the litigation to me. A straightforward, level-voiced man, he had got his start as a mining engineer and then come back home with his shirtsleeves rolled up, ready for a lengthy bout of legal wrangling. After multiple trials and appeals, the Supreme Court of Canada ruled in 1997 that British Columbia had to negotiate the status of the land with the Gitksan. Talks were still ongoing in 2005, two decades after the lawsuit first began.

After a while Sterritt took me to see 'Ksan, a historical park and art school

created in 1970. In the park were several re-created longhouses, their facades covered in the forcefully elegant, black-and-red arcs of Northwest Coast Indian art. The art school trained local Indians in the techniques of translating traditionally derived designs into silk-screen prints. Sterritt left me in a back room of the schoolhouse and told me to look around. There was more in the room than he may have realized, for I quickly found what looked like storage boxes for a number of old and beautiful masks. Beside them was a stack of modern prints, some of which used the same designs. And there were boxes of photographs, old and new alike, many of splendid artworks.

In Northwest Coast art the subjects are flattened and distorted—it's as if they've been reduced from three dimensions to two and then folded like origami. At first I found all the designs hard to interpret, but soon some seemed to pop right out of the surface. They had clean lines that cut space into shapes at once simple and complex: objects tucked into objects, creatures stuffed into their own eyes, humans who were half beast and beasts who were half human—all was metamorphosis and surreal commotion.

A few of the objects I looked at I understood immediately, many I didn't understand at all, some I thought I understood but probably didn't, and some maybe even the Gitksan didn't understand, in the way that most Europeans today can't truly understand the effect of Byzantine art on the spirits of the people who saw it at the time of its creation. But I was delighted by the boldly graphic lines and dazzled by the sense that I was peeking into a vibrant past that I had not known existed and that continued to inform the present in a way I had not realized. For an hour or two I went from object to object, always eager to see more. In assembling this book, I hope to share the excitement I felt then, and have felt many times since.

INTRODUCTION

Holmberg's Mistake



Accompanying him were twelve workers from the nearby hamlet of Tres Zapotes. They pried the stela from the ground with wooden poles, but it was blank. Disappointed, Stirling took the crew to yet a third fallen stela. They scraped away the covering dirt and found that it, like the first, was covered with intricate images. Alas, the carvings were now too weathered to be deciphered. The frustrated Stirling asked the workers to expose the back of the slab by digging beneath it and levering up the stone with poles. Several men, he later recounted, "were on their knees in the excavation, cleaning the mud from the stone with their hands, when one of them spoke up in Spanish: 'Chief! Here are numbers!'"

Across the back of the stela were clumps of dots and bars, a notation familiar to Stirling from the Maya. The Maya used a dot to signify one and a horizontal bar to signify five; the number nineteen would thus be three bars and four dots. Stirling copied the dots and bars and "hurried back to camp where we settled down to decipher them." The inscription turned out to be a date: September 3, 32 B.C., in today's calendar.

Stirling already knew that Tres Zapotes was anomalous—it was at least 150 miles west of any previously discovered Maya settlement. The date deepened the puzzle. If, as seemed likely, it recorded when the stela was put on display, this implied that Tres Zapotes had been a going concern in 32 B.C.—centuries before any other known Maya site. The date thus seemed to imply that the Maya had originated well to the west of what was thought of as the traditional homeland, and much earlier than had been thought. Stirling didn't believe it. Surely the Maya had not sprung up in Tres Zapotes and then moved en masse hundreds of miles to the east. But the alternative explanation—that Tres Zapotes was not a Maya community—seemed equally improbable. The Maya were universally regarded as the oldest advanced society in Mesoamerica. Whoever had carved the stela had some knowledge of writing and mathematics. If they were not Maya, the implication was that someone else had launched the project of civilization in Mesoamerica.

Learning from local people that Tres Zapotes was only one of many mound sites in Veracruz, Stirling decided to return in 1940 to survey them all. The task was daunting even for a cigar-chomping, whisky-drinking, adventure addict like Stirling. Most of the mound centers were in the middle of trackless mangrove swamps or up narrow, unmapped rivers choked with water hyacinth. Ticks and mosquitoes were indefatigable and present in huge numbers; the ticks were worse than the mosquitoes, Stirling remarked because they had to be dug out of the flesh with a knife. At one point Stirling and a colleague hitched a ride in a pepper truck to one of the smaller sites. After jolting down a road with deep ruts "designed to test the very souls of



Writing, Wheels, and Bucket Brigades

(*Tales of Two Civilizations, Part II*)

"LIKE GRAPES THEY FALL OFF"

On January 16, 1939, Matthew W. Stirling took an early-morning walk through the wet, buggy forest of Veracruz state, on the Gulf Coast side of Mexico's southern isthmus. Eighty years before his walk, a villager tramping through the same woods had stumbled across a buried, six-foot-tall stone sculpture of a human head. Although the find was of obvious archaeological importance, the object was so big and heavy that in the intervening eight decades it had never been pulled out of the ground. Stirling, director of the Smithsonian Bureau of American Ethnology, had gone to Mexico the year before, in early 1938, to see the head for himself. He found it, sunk to the eyebrows in mud, after an eight-hour horseback ride from the nearest town. The head was in the midst of about fifty large, artificial earthen mounds—the ruins, Stirling concluded with excitement, of a previously unknown Maya civic center. He had decided to assemble a research team and explore the area in more detail the next year, and persuaded the National Geographic Society to foot the bill. When he returned to Veracruz, he and his team cleared the dirt around the great head, admiring its fine, naturalistic workmanship, so unlike the stiff, stylized sculpture common elsewhere in Mesoamerica. Nearby, they found a stela, its wide, flat face covered with bas-relief figures. Hoping to turn up others, Stirling was walking that January morning to the far end of the mounded area, where a workman had noticed a large, flat, partly submerged rock: a second stela.

motorcars," the two men were let off in a nondescript meadow. Stirling went to talk with the driver.

"The ticks are not bad, are they?" I asked him hopefully, viewing the tall grass and underbrush between the road and the mounds. "No," said the driver, beaming. "When full, like grapes they fall off and no harm is done. There are millions of them here, however."

In La Venta, a dry, raised "island" in the coastal swamp, Stirling's team discovered four more colossal heads. Like the first, they had no necks or bodies and wore helmets that vaguely resembled athletic gear. All were at least six feet tall and fifteen feet round and made from single blocks of volcanic basalt. How, Stirling wondered, had their makers transported these ten-ton blocks from the mountains and across the swamp? Whoever these people were, he eventually concluded, they could not be Maya; their ways of life seemed too different. Instead they must have belonged to another culture altogether. La Venta was filled with mounds and terraces, which told Stirling that many people had lived there. The city, he wrote in 1940, "may well be the



Curious villagers surround the great Olmec head excavated in 1939 by archaeologist Matthew Stirling in the Mexican state of Veracruz.

basic civilization out of which developed such high art centers as those of the Maya, Zapotecs, Toltecs, and Totonacs." He called its "mysterious people" the Olmec.

Stirling's account set the template for decades to follow. Ever since his day, the Olmec have been known by two Homeric epithets: they were "mysterious," and they were the "mother culture" of Mesoamerica. (Tourists are told by Frommer's 2005 Mexico guide, for example, to visit the ruins of the "enigmatic people" who created the "mother culture of Mesoamerica.") But in recent years many archaeologists have come to believe that neither description is correct.

The Olmec's purported mysteriousness is related to their emergence. To Stirling and many of his successors, the Olmec seemed to have no peers or ancestors; they appeared fully formed, apparently from nowhere, like Athena springing from the brow of Zeus. First there was a jungle with a few indistinguishable villages; then, suddenly, a sophisticated empire with monumental architecture, carved stelae, earthwork pyramids, hieroglyphic writing, ball courts, and fine artworks—all of it conjured into existence with the suddenness of a magician's trick. The Olmec, wrote Smithsonian archaeologist Betty Meggers, were a "quantum change." Their status as precursors led archaeologists to believe that the subsequent emergence of other complex societies was due to their example—or their conquest. Even the mighty Maya did little more than continue down the path set by the Olmec. "There is now little doubt," Yale archaeologist Michael Coe wrote in 1994, "that all later civilizations in Mesoamerica, whether Mexican or Maya, ultimately rest on an Olmec base."

Strictly speaking, Coe was mistaken. By the time he wrote, many of his colleagues strongly doubted that the Olmec either emerged alone or were the mother culture. They did emerge abruptly, these researchers say, but they were only the first of the half-dozen complex societies—"sister cultures"—that sprang up in southern Mexico after the development of maize agriculture. Focusing on the Olmec's chronological primacy, they believe, obscures the more important fact that Mesoamerica was the home of a remarkable multisociety ferment of social, aesthetic, and technical innovation.

RUBBER PEOPLE

Nobody knows the right name for the Olmec, but "Olmec" is the wrong one. They spoke a language in the Mixe-Zoquean language family, some members of which are still used in isolated pockets of southern Mexico. "Olmec," though, is a word in Nahuatl, the language of the Mexica to the north. It

means, more or less, "people of the land of rubber." The problem with the name is not so much that the Olmec did not use it for themselves—nobody knows what that name was, and they have to be called something. Nor is the problem the rubber, which the Olmec used, and may have invented (scientists discovered in the 1990s that they made rubber by chemically treating the latex-containing sap of a tropical tree, *Castilla elastica*). The problem is that the Mexica did not actually use the name to refer to the putative mother culture in Veracruz, but to another, completely unrelated culture in Puebla to the west, a culture that, unlike the ancient Olmec, still existed at the time of the Spanish conquest. The confusion between the Mexica's Olmec and Stirling's Olmec led some archaeologists to propose that the latter should be called the "La Venta Culture," after the site he investigated. Almost everyone agreed that the new name was a big improvement, logically speaking. Unfortunately, nobody used it. Not for the first time in Native American history, the confusing, incorrect name prevailed.

The Olmec heartland was the coastal forests of Veracruz. Compared to the Norte Chico, the area is promising. Like the Peruvian littoral, it is bracketed by sea and mountains, but it catches, rather than misses, the prevailing winds, and the rain that comes with them. The shoreline itself is swampy, but not far from the coast the country rises into a lush, fertile plateau. Further inland are the Tuxtla Mountains, with many rivers cascading down their flanks. The rivers flood in the rainy season, enriching the land, Nile Delta style. During the rest of the year, the climate is drier, and farmers plant and tend their *milpas* on the alluvial soil.

The first traces of the people who would become the Olmec date back to about 1800 B.C. At that time there was little to distinguish them from groups elsewhere in Mesoamerica. But something happened in Veracruz, some spark or incitement, a cultural quickening, because within the next three centuries the Olmec had built and occupied San Lorenzo, the first large-scale settlement in North America—it covered 2.7 square miles. On a plateau commanding the Coatzacoalcos river basin, San Lorenzo proper was inhabited mainly by the elite; everyone else lived in the farm villages around it. The ceremonial center of the city—a series of courtyards and low mounds, the latter probably topped with thatch houses—sat on a raised platform 150 feet high and two-thirds of a mile to a side. The platform was built of almost three million cubic yards of rock, much of it transported from mountain quarries fifty miles away.

Scattered around the San Lorenzo platform were stone monuments: massive thrones for living kings, huge stone heads for dead ones. Rulers helped to mediate between supernatural forces in the air above and the

watery place below where souls went after life. When kings died, their thrones were sometimes transformed into memorials for their occupants: the colossal heads. The features of these enormous portraits are naturalistically carved and amazingly expressive—thoughtful or fiercely proud, mirthful or dismayed. It is assumed they were placed like so many stone sentinels for maximum Orwellian impact: the king is here, the king is watching you.*

Like the carvings and stained-glass windows in European cathedrals, the art in San Lorenzo and other Olmec cities consisted mainly of powerful, recurring images—the crucifixions and virgins, so to speak, of ancient Mesoamerica. Among these repeated subjects is a crouched, blobby figure with a monstrously swollen head. Puzzled researchers long described these sculptures as "dwarves" or "dancers." In 1997 an archaeologist and a medical doctor with archaeological leanings identified them as human fetuses. Their features were portrayed accurately enough to identify their stage of development. Researchers had not recognized them because artistic renditions of fetuses are almost unheard of in European cultures (the first known drawing of one is by Leonardo). Other frequent themes included lepers, the pathologically obese, and people with thyroid deficiencies, all portrayed with a cool eye for anatomical detail. Perhaps the best-known subject is a man or boy gingerly holding a "were-jaguar": a limp, fat, sexless baby with a flattened nose and a snarling jaguar mouth. Often the baby has a deeply cloven skull.

The denizens of San Lorenzo are unlikely to have shared Europeans' dismay at the physical deformity portrayed in these images. Indeed, by contemporary standards high-born Olmec were deformed themselves. By binding small, flat pieces of wood to newborns' foreheads, they pushed up the soft infant bones, making the skull longer and higher than normal. To further proclaim their status, wealthy Olmec carved deep grooves into their teeth and pierced their nasal septums with bone awls, plugging the holes with ornamental jade beads. (Because no Olmec skeletons have been found, no direct proof of these practices exists; instead archaeologists base their beliefs on the portrayal of Olmec nobles in figurines and sculptures.)

Swanning about the elite precincts, the rich and powerful wore finely

*The statues' broad lips and flat noses have led "Africanist" historians like Clyde Winters and Ivan Van Sertima to claim that the Olmecs either were visited by Africans or had actually migrated from Africa. The African knowledge gained thereby explains the Olmec's rapid rise. These views are not widely endorsed. Surprisingly, several noted archaeologists, including Betty Meggers and Gordon Ekholm, have suggested the geographical opposite: that Olmec society was inspired by China. Visitors from the Shang Dynasty are said to have crossed the Pacific to teach the ancient Olmec how to write, build monuments, and worship a feline god. This hypothesis, too, has failed to stir enthusiasm.

woven clothing, but only below the waist—breachcloths for men, skirts and belts for women. Veracruz was too hot for anything more. On public occasions, nobles bedizened themselves with bracelets, anklets, many-stranded necklaces, bejeweled turbans, and big, hip-hop-style pendants. Some of the last were concave mirrors made from beautifully polished magnetite. Precisely ground, the mirrors were able to start fires and project images onto flat surfaces, *camera lucida* fashion. Presumably they were used to dazzle hoi polloi. As for the poor, it is likely that they went naked, except possibly for sandals.

San Lorenzo fell in about 1200 B.C., victim of either revolution or invasion. Or perhaps it was abandoned and sacked for religious reasons—archaeologists have advanced several hypotheses for the city's demise. What is certain is that the site was vacated and the stelae defaced and the sculptures decapitated. The colossal heads being, so to speak, pre-decapitated, they were smashed with hammers and systematically buried in long lines. Vegetation overran the red-ocher floors and the workshops that manufactured ceramic figurines and iron beads and rubber ax-head straps.

Olmec society was surprisingly unaffected by the collapse of its greatest polity. A much bigger city, La Venta, was going up on a swamp island about forty miles away.

Today La Venta is partly buried by an oil refinery, but in its heyday—roughly speaking, 1150 B.C. to 500 B.C.—it was a large community with a ring of housing that surrounded a grand ceremonial center. The city's focus, its Eiffel Tower or Tiananmen Square, was a 103-foot-tall clay mound, a bulging, vertically fluted cone somewhat resembling a head of garlic. The mound rose at the south end of a rectangular, hundred-yard-long pavilion that was bordered by two knee-high berms. At the north end of the pavilion was a sunken rectangular courtyard fenced on three sides by a row of seven-foot basalt columns atop a low red and yellow adobe wall. The fourth, northern side opened onto a third mound, larger than the small mound but nowhere near the size of the big one. The pavilion and courtyard had painted walls and floors of colored sand and clay, heavy sculptural objects, including several of the trademark heads, studded the area. This central part of the city was reserved, archaeologists believe, for clerics and rulers. It was Buckingham Palace and the Vatican rolled into one.

La Venta, too, was destroyed, perhaps deliberately, around 350 B.C. But its eight-hundred-year existence spanned one of the most exciting times in American history. At La Venta's height, Olmec art and technical innovations could be found throughout Mesoamerica. So widespread was Olmec iconography—jaguar babies, carved stelae, distinctively shaped ceramics—that many archaeologists believed its very ubiquity was evidence that the Olmec

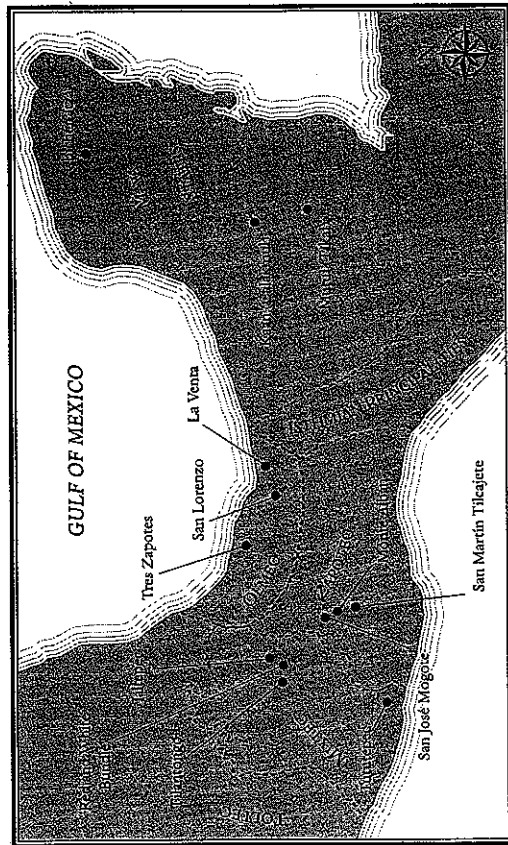
"not only engendered Mesoamerica but also brought forth the first Mesoamerican empire." The description is from Ignacio Bernal, once director of Mexico's famed National Museum of Anthropology. Bernal, who died in 1992, envisioned an imperium that spanned much of southern Mexico, proselytizing its religion and forcing other groups to send their finest works to its heartland. The Olmec, he thought, were the Romans of Mesoamerica, a magisterial society that "established the pattern which, through the centuries, was to be followed by other expansionist Mesoamerican cultures."

Since Bernal's death many researchers have come to view the Olmec differently. According to the University of Michigan anthropologists Kent Flannery and Joyce Marcus, the Olmec heartland was but one of four regional power centers: the Central Basin to the north, where settlements like Tlatilco and Tlapacoya laid the groundwork for the empires of Teotihuacán and the Toltecs; in the isthmus, the chiefdoms of Oaxaca; the Olmec, along the Gulf Coast; and, later, the Maya polities in Yucatán and northern Guatemala. Some believe that there was a fifth power: Chalcatzingo, an important chiefdom between the Central Basin and Oaxaca.

In the first millennium B.C., all four (or five) were making the transition from individual fortified villages to groups of chiefdoms to states with centralized authority. Although the Olmec preceded the others, they did not set the template for them. Instead they all influenced each other, sometimes by trade, sometimes by violence, each one developing new techniques, exporting unique goods, and swiping ideas from the others. In this world of "competitive interaction," all parties hustled for advantage. Trade in goods was important, but it was the trade in ideas that mattered.

By making this argument, I am endorsing one side in the long-running dispute between mother- and sister-culture proponents. In other words, as one mother-culture advocate put it, I am "swallowing Marcus's [nonsense] whole." He may be right. But I would argue that there is a difference between *inheriting* a cultural tradition, as the Norte Chico culture's successors apparently did, and *copying* one, as the Olmec's proponents argue. Nobody disputes that Han Chinese society arose long before its neighbors in Asia, but Asian archaeologists don't refer to China as Asia's "mother culture," because China's neighbors used part of its intellectual legacy to build up their own distinct and different writing systems, agricultural technologies, imperial practices, and much else. Given the evidence available now, the same seems to be true for the Olmec and ancient Mesoamerica. (To me, anyway—many researchers disagree.)

Emblematic of this rocketing growth were the Olmec's neighbors in Oaxaca, the Zapotecs, whom Flannery and Marcus have studied for more than two decades. The Zapotecs were based across the mountains from the



MESOAMERICA, 1000 B.C.-1000 A.D.

For thousands of years Mesoamerica was a wellspring of cultural innovation and growth. This map does not depict it accurately, because these societies were not contemporaries—the Olmec vanished centuries before the Mixtec and Zapotec began to reach their height, for example. Nonetheless, it should give some idea of each society's heartland.

Olmec, in Oaxaca's high Central Valley—three forty-to-sixty-mile-long bowls that intersect in a ragged Y. By about 1550 B.C., they were abandoning the life of hunting and gathering to live in villages with defensive palisades.* These early villages had wattle-and-daub houses, fine pottery, and some public architecture. They were controlled by "big men," the social scientist's term for the alpha male who is able in such informal settings to enforce his will through persuasion or force. Within a few hundred years, the big men acquired *rank*—that is, they began to wield power not only because of their personal charisma, but also because their societies had given them an elevated official position. It is the difference between the leading spirit on a pickup basketball team and the officially selected captain of a college squad. After this, political consolidation proceeded apace. Soon the valley was dominated by three main chiefdoms, one at each endpoint of the Y. They did not get along; a thirty-square-mile buffer zone in the middle, Flannery and Marcus noted, was "virtually unoccupied."

*Here, as elsewhere in this book, I am being chronologically inexact. The oldest Zapotec palisade Flannery and Marcus excavated yielded calibrated radiocarbon dates in the range between 1680 and 1410 B.C., which for brevity's sake I render as "about 1550 B.C."

The biggest of the three chiefdoms was located near today's village of San José Mogote. Around 750 B.C. it was attacked and its temple razed in a fire so intense that the clay walls melted. San José Mogote quickly rebuilt the temple a few yards away.

Across the new threshold artisans laid a carved stone. Splayed across its face, suitable for stomping on, was a bas-relief of a naked, disemboweled male corpse, apparently a defeated enemy, blood welling from his side. The carving, to my taste, is beautifully done, as graphically elegant, despite the gory subject matter, as a Matisse. What matters most to researchers, though, is not the design but the two curious marks between the cadaver's feet. The two marks are glyphs: the oldest firmly dated writing in the Americas. Despite their brevity, they are full of information, both because of the nature of written language, and because of the special way Mesoamerican people chose their names.

COUNTING AND WRITING

Writing begins with counting. When a culture grows big enough, it acquires an elite, which needs to monitor things it considers important: money, stored goods, births and deaths, the progression of time. In Mesopotamia, Sumerian accountants began keeping records with clay tokens around 9000 B.C. As the need for precision grew, they scratched marks on the tokens as mnemonic devices. For example, they might have distinguished a count of sheep from one of wheat by drawing a sheep on one and a wheat stalk on the other. Gradually the information on each record increased. The bureaucrats were not intending to create writing. Instead they were simply adding useful features as they became necessary. By 3200 B.C. Sumerian scribes had progressed to inscribing on clay tablets with sharpened reeds. A tablet might contain, say, two hash marks, a box, a circle with a cross in the middle, an asterisk-like shape, and an arrangement of three triangles. Scribes would know that the hash marks meant "two," the box was a "temple," the circle stood for "cattle," an asterisk meant "goddess," and the triangles were "Inanna"—two cattle owned by the goddess Inanna's temple. (Here I am lifting an example from Gary Urton, a Harvard anthropologist.) They had no way to indicate verbs or adjectives, no way to distinguish subject from object, and only a limited vocabulary. Nonetheless, Sumerians were moving toward something like writing.

In Mesoamerica, timekeeping provided the stimulus that accounting gave to the Middle East. Like contemporary astrologers, the Olmec, Maya, and Zapotec believed that celestial phenomena like the phases of the moon and

Venus affect daily life. To measure and predict these portents requires careful sky watching and a calendar. Strikingly, Mesoamerican societies developed three calendars: a 365-day secular calendar like the contemporary calendar; a 260-day sacred calendar that was like no other calendar on earth; and the equally unique Long Count, a one-by-one tally of the days since a fixed starting point thousands of years ago. Establishing these three calendars required advances in astronomy; synchronizing them required ventures into mathematics.

The 260-day ritual calendar may have been linked to the orbit of Venus; the 365-day calendar, of course, tracked the earth's orbit around the sun. Dates were typically given in both notations. For example, October 12, 2004, is 2 Lamat 11 Yax, where 2 Lamat is the date in the ritual calendar and 11 Yax the date in the secular calendar. Because the two calendars do not have the same number of days, they are not synchronized; the next time 2 Lamat occurs in the sacred calendar, it will be paired with a different day in the secular calendar. After October 12, 2004, in fact, 2 Lamat and 11 Yax will not coincide again for another 18,980 days, about fifty-two years.

Mesoamerican cultures understood all this, and realized that by citing dates with both calendars they were able to identify every day in this fifty-two-year period uniquely. What they couldn't do was distinguish one fifty-two-year period from another. It was as if the Christian calendar referred to the year only as, say, '04—one would then be unable to distinguish between 1904, 2004, and 2104. To prevent confusion, Mesoamerican societies created the third calendar, the Long Count. The Long Count tracks time from a starting point, much as the Christian calendar begins with the purported birth date of Christ. The starting point is generally calculated to have been August 13, 3114, B.C., though some archaeologists put the proper date at August 10 or 11, or even September 6. Either way, Long Count dates consisted of the number of days, 20-day "months," 360-day "years," 7,200-day "decades," and 144,000-day "millennia" since the starting point. Archaeologists generally render these as a series of five numbers separated by dots, in the manner of Internet Protocol addresses. Using the August 13 starting date, October 12, 2006, would be written in the Long Count as 12.19.13.12.18. (For a more complete explanation, see Appendix D.)

Because it runs directly from 1 B.C. to 1 A.D., the Christian calendar was long a headache for astronomers. Scientists tracking supernovae, cometary orbits, and other celestial phenomena would still have to add or subtract a year manually when they crossed the A.D.-B.C. barrier if a sixteenth-century astronomer named Joseph Scaliger hadn't got sick of the whole business and devised a calendar for astronomers that doesn't skip a year. The Julian calen-

dar, which Scaliger named after his father, counts the days since Day 0. Scaliger chose Day 0 as January 1, 4713, B.C.; Day 1 was January 2. In this system, October 12, 2006, is Julian Day 2,454,021.

The Long Count calendar began with the date 0.0.0.0.0.* Mathematically, what is most striking about this date is that the zeroes are true zeroes. Zero has two functions. It is a number, manipulated like other numbers, which means that it is differentiated from nothing. And it is a placeholder in a positional notation system, such as our base-10 system, in which a number like 1 can signify a single unit if it is in the digits column or ten units if it is in the adjacent column.

That zero is not the same as nothing is a concept that baffled Europeans as late as the Renaissance. How can you calculate with nothing? they asked. Regarding zero as a dangerous idea, the Catholic Church banned Hindu-Arabic numerals—the 0 through 9 used today—in much of Italy until the fourteenth century. A classic demonstration of zero's status as a number, according to science historian Dick Teresi, is grade point average:

In a four-point system, an A equals 4, B equals 3, and so on, down to E, which equals 0. If a student takes four courses and gets A's in two but fails the other two, he receives a GPA of 2.0, or a C average. The two zeroes drag down the two A's. If zero were nothing, the student could claim that the grades for the courses he failed did not exist, and demand a 4.0 average. His dean would laugh at such logic.

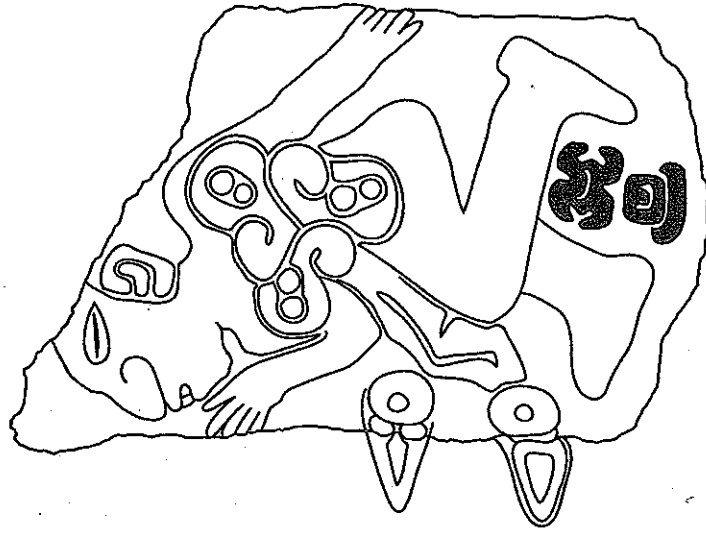
Without a positional notation system, arithmetic is tedious and hard, as schoolchildren learn when teachers force them to multiply or subtract with Roman numerals. In Roman numerals, CLIV is 154, whereas XLII is 42. Mad-denigly, both numbers have L (50) as the second symbol, but the two L's aren't equivalent, because the second is modified by the preceding X, which subtracts ten from it to make forty. Even though both CLIV and XLII are four-digit numbers, the left-hand symbol in the first number (C) cannot be directly compared with the left-hand symbol in the second (X). Positional notation symbols take the aggravation out of arithmetic.

*Actually, it didn't. Inexplicably, the biggest unit, the 144,000-day "millennium," began with 1, rather than 0. The first day in the calendar was thus 1.0.0.0.0. When I remarked on the peculiarity of this exception to a mathematician, he pointed out societies whose timekeeping systems are so irregular that children have to learn rhymes to remember the number of days in the months ("Thirty days hath September . . .") are in no position to scoff at the calendrical eccentricities of other cultures. At least all the "months" in the Mesoamerican calendar had the same number of days, he said.

Stirling's stela in Tres Zapotes bore a Long Count date of 7.16.6.16.18. The implication is that by 32 B.C. the Olmec already had all three calendars and zero to boot. One can't be sure, because the date does not include a zero or a reference to the other calendars. But it is hard to imagine how one could have a Long Count without them. Tentatively, therefore, archaeologists assign the invention of zero to sometime before 32 B.C., centuries ahead of its invention in India.

How long before 32 B.C.? The carved cadaver in San José Mogote may give a hint. In Mesoamerican cultures, the date of one's birth was such an important augury of the future that people often acquired that day as their name. It was as if coming into the world on New Year's Day were such a sign of good fortune that children born on that day would be named "January 1." This seems to have been the case for the man whose death was celebrated in the San José Mogote temple. Between his feet are two glyphs, one resembling a stovepipe hat with a U painted across the front, the other looking vaguely like a smiling pet monster from a Japanese cartoon. According to Marcus, the Michigan anthropologist, the glyphs correspond to 1-Earthquake, the Zapotec name for the seventeenth day of the 260-day sacred calendar. Because the carving depicts a man instead of an event, the date is generally thought to be the dead man's name. If so, 1-Earthquake is the first named person in the history of the Americas. Even if the date is not a name, the two glyphs indicate that by 750 B.C., when the slab was carved, the Zapotec were not only on the way to some form of writing, but had also assembled some of the astronomical and mathematical knowledge necessary for a calendar.

To judge by the archaeological record, this development took place in an astonishingly compressed period; what took the Sumerians six thousand years apparently occurred in Mesoamerica in fewer than a thousand. Indeed, Mesoamerican societies during that time created more than a dozen systems of writing, some of which are known only from a single brief text. The exact chronology of their evolution remains unknown, but could be resolved by the next object that a farmer discovers in a field. The earliest known Olmec writing, for example, is on a potsherd from Chiapas that dates from about 300 B.C. For a long time nobody could read it. In 1986 a workcrew building a dock on the Acuña River in Veracruz pulled out a seven-foot stela covered with Olmec symbols. Thought to have been written in 159 A.D., the twenty-one columns of glyphs were the first Olmec text long enough to permit linguists to decipher the language. Two linguists did just that in 1993. The stela recounted the rise of a warrior-king named Harvest Mountain Lord who celebrated his ascension to the throne by decapitating his main rival during the



Discovered in 1975, this prone, disemboweled man was carved onto the stone threshold of a temple in San José Mogote, near the city of Oaxaca. Between the corpse's feet is the oldest certainly dated writing in the Americas: two glyphs (shaded in drawing) that probably represent his name, 1-Earthquake. The ornate scroll issuing from his side is blood. According to Joyce Marcus, the first archaeologist to examine this bas-relief, the Zapotec words for "flower" and "sacrificial object" are similar enough that the flowery blood may be a visual pun.

This information in hand, the linguists went back to the writing on the potsherd. Disappointingly, it turned out to be some banal utterances about dying and cutting cloth.

FROM 1-EARTHQUAKE TO 8-DEER

The development of writing in Zapotec society went hand-in-hand with growing urbanization. In about 500 B.C., San José Mogote seems to have transplanted itself to Monte Albán, in the middle of the buffer zone. About

half an hour by bus from Oaxaca City, Monte Albán is today a decorous sprawl of walls and pyramids enveloped by a lush lawn (this last is an import from Europe; lawn grass did not exist in the Americas prior to Columbus). Arriving tourists are hailed by "guides" with backpacks full of phony ancient figurines and ethnically incorrect souvenirs of Mexico drawings. Their ministrations do not diminish the lonely dignity of the ruins. Monte Albán is atop a steep, 1,500-foot hill that overlooks the valley of Oaxaca. The Zapotec reconfigured the entire hill to build the city, slicing out terraces and platforms. By leveling the entire summit, they created a fifty-five-acre terrace half the size of the Vatican. At its zenith, Monte Albán housed seventeen thousand people and was by a considerable margin the biggest and most powerful population center in Mesoamerica.

The rationale for its construction is the subject of yet another lengthy archaeological dispute. One side proposes that Monte Albán formed because maize agriculture allowed the Oaxaca Valley's population to grow so much that the rural villages naturally clustered into something resembling cities. For most of its history Monte Albán was thus a huge village, not a true city, and certainly not a hierarchical state. Others argue that warfare had grown so devastating, as shown by the destruction of San José Mogote, that the main valley chiefdoms formed a defensive confederation headquartered at Monte Albán. Yet a third theory is that the Zapotec of Monte Albán—not the Olmec of La Venta—consolidated to form North America's first imperialist power, an aggressive state that subjugated dozens of other villages.

Among the strongest evidence for the last view are the nearly three hundred carved stone slabs at Monte Albán that depict slain, mutilated enemies: the rulers, Marcus believes, of communities conquered by Monte Albán. Some of the stones are labeled with enemy names, as with the unfortunate 1-Earthquake. These may commemorate victories in Monte Albán's grinding battle for supremacy with its local rival, San Martín Tilcajete, in the southern arm of the Central Valley. When San José Mogote founded Monte Albán, Tilcajete responded by gathering people from its surrounding villages, doubling in size, and erecting its own ceremonial buildings. War was the inevitable result. Monte Albán sacked Tilcajete in about 375 B.C. Undiscouraged, Tilcajete rebuilt itself on a better defensive position and acquired larger armies. When it again became a threat, Monte Albán attacked for the second time in 120 B.C. This time its forces finished the job. They burned the king's palace to the ground and emptied the rest of Tilcajete, leaving Monte Albán firmly in control of the entire valley.

With nothing to impede it, Monte Albán swept out and established a domain of almost ten thousand square miles. For centuries it stood on equal ground with its neighbors, the rising Maya states to the east and Teotihuacán

to the north. It enjoyed relatively peaceful relations with both but had continual trouble in Mixteca, a constellation of petty principalities immediately to the west. By contrast with Monte Albán, these were minuscule entities; most were clusters of rustic villages covering ten to twenty square miles. Yet they were amazingly troublesome. Monte Albán repeatedly overran the Mixtec statelets, but never managed to eliminate them. These tiny, fractious domains endured for more than a thousand years. Meanwhile the much stronger and more centralized Zapotec empire collapsed completely in about 800 A.D.

Mixtec writing survives in eight codices, the deerskin or bark "books" whose painted pages could be folded like screens or hung on the wall like a mural. (The Spaniards destroyed all the rest.) More purely pictorial than Zapotec or Maya script, the texts were arranged almost randomly on the page; red lines directed the reader's eye from image to image. The symbols included drawings of events, portraits labeled by name (the king 4-Wind, for example, being shown by symbolic wind and four little bubbles in a line), and even punning rebuses. Enough writing has survived to give, when coupled with archaeological studies, a vivid picture of Mixtec life.

Like medieval Italian city-states, Mixtec principalities were rigidly stratified; with the king and a small group of kinspeople and noble advisers gobbling up much of the wealth and land. They constantly shifted configuration, some expanding by swarming over their neighbors, others imploding when their constituent villages seceded and joined other polities. More commonly, two states joined when their rulers married. Alliance through royal marriage was as common in eleventh-century Mixteca as it was in seventeenth-century Europe. In both, royal family trees formed an intricate network across national boundaries, but in Mixteca the queen's lands stayed in her line—the king's heir wasn't necessarily the queen's heir. Another difference: primogeniture was not expected. If the queen did not think her eldest son was fit for the crown, she could pass it to another child, or even to a nephew or cousin.

No fewer than four of the codices treat the story of 8-Deer Jaguar Claw, a wily priest-general-politician with a tragic love for the wife of his greatest enemy. Born in 1063 A.D., 8-Deer was a shirttail cousin to the ruling family of Tilantongo, which had been engaged for decades in a dynastic struggle with the kingdom of Red and White Bundle. (The name, a modern invention, comes from its name-glyph, which pictures the cloth wrapping used by the Mixtec to wrap holy objects; its exact location is still not nailed down.) Like his father, a high cleric, 8-Deer was trained for the priesthood, but political events and his own overweening ambition stopped him from following that path.

After an unprovoked attack on Red and White Bundle by Tilantongo

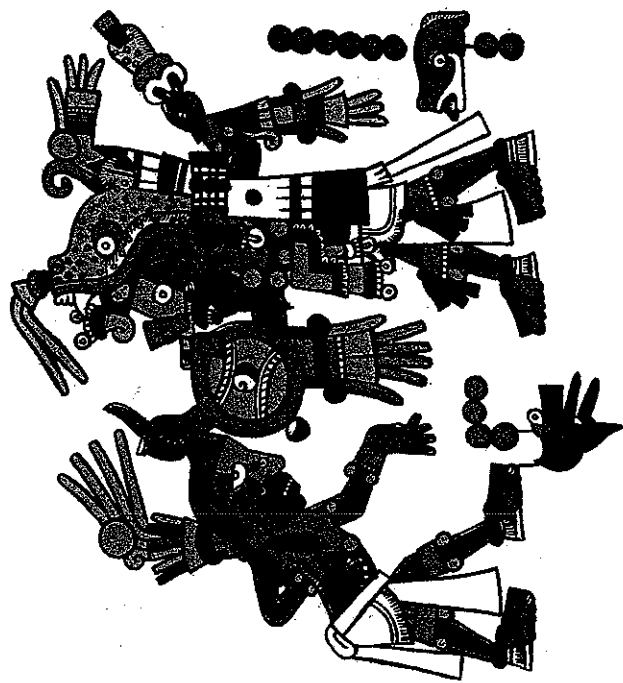
raised hostilities to a fever, the warring parties agreed to meet in a sacred mountain cave with the Priestess of the Dead, a powerful oracle who had stripped away the flesh from her jaw, giving her a terrifying, skull-like appearance. Tilantongo's representative was 8-Deer, who attended the meeting in place of his recently deceased father. To his dismay, the priestess sided with Tilantongo's enemies and ordered 8-Deer, Tilantongo's champion, to exile himself a hundred miles away, in a jerkwater town on the Pacific called Tututepec.

Tucked away in Tututepec, 8-Deer assembled a private army, staffed it with many relatives, and in a series of swift campaigns seized dozens of neighboring villages and city-states. In addition to assembling the greatest empire ever seen in the region, the conquests managed to kill off most of the siblings and cousins above him in the line of royal succession. After six years of war he returned home to Tilantongo. During this visit, according to John M. D. Pohl, the archaeologist whose interpretations I am mostly following here, 8-Deer accidentally encountered 6-Monkey, the young wife of the much older king of Red and White Bundle. Despite the long enmity between the two kingdoms, 8-Deer and 6-Monkey secretly became lovers.

In 1096 Tilantongo's sovereign died in mysterious circumstances. The Priestess of the Dead selected 8-Deer's beloved elder half brother to be the regent—that is, the half brother became the last person between 8-Deer and the throne of Tilantongo. Three years later, unknown assailants stabbed the half brother to death in a sweatbath. The inconsolable 8-Deer took the throne of Tilantongo and declared war on Red and White Bundle, which he claimed had orchestrated the murder.

Red and White Bundle's royal palace was built on a cliff over a bend in the river. Guarded by sheer walls on three sides, its soldiers had only to watch the fourth side, across which was an earthen berm. Leading an army of a thousand, 8-Deer threw up ladders, swarmed over the berm with his men, and entered the palace. As befit a conqueror, 8-Deer was wearing elaborate cotton armor, a ceremonial beard wig, and a cowl made from the head of a jaguar. Gold-and-jade necklaces dangled across his naked chest. In the palace he found 6-Monkey and her husband, the king of Red and White Bundle. Both were mortally wounded. In Pohl's account, 8-Deer held 6-Monkey as she died.

Captured with the royal couple were their two sons, the elder of whom, 4-Wind, was heir to the throne. Seizing him by the hair, 8-Deer forced the teenager to grovel before him. But he also made what seems to have been a sentimental decision: he spared the life of his lover's son. The folly of this action became apparent when 4-Wind and his brother escaped from confinement.



In this fragment from a Nudzahui codex, the jaguar-cowled Lord 8-Deer (right) captures 4-Wind, son of his former lover, by the hair. As in other Nudzahui codices, the characters' names are indicated by the accompanying circles-plus-head symbols.

Seeking revenge, 4-Wind approached the Zapotec empire for help. With Zapotec backing, he linked rebels in Red and White Bundle and a host of other cities defeated by 8-Deer. They besieged Tilantongo in 1115. The battle lasted six months and ended in total defeat for Tilantongo. In a mirror image of the past, the captured 8-Deer was forced to bow to 4-Wind. He was fifty-five years old and had six official kingships and dozens of petty states under his control. Victorious and vengeful, 4-Wind personally disemboweled him. Then he married 8-Deer's daughter.

In 4-Wind's first exercise of statecraft, he abandoned the Zapotec allies who had helped him achieve the throne, aligned Tilantongo with the Toltec empire to the north, and attacked the Zapotec. Ultimately the Nudzahui under his lead took over much of Oaxaca, forcing the Zapotec states to pay tribute. The empire he established, far bigger than 8-Deer's, lasted until the fifteenth century, when the Mexica invaded. And then came Cortés.

WHEELED INTERLUDE

As Matthew Stirling and his team were dodging ticks and unearthing stelae in Tres Zapotes, they found a cache of fifteen upside-down pottery bowls tucked into the ground six feet below the surface. The bowls protectively covered thirty-five toy-size, decorated figurines and twelve small, painted clay discs. Among the figurines were two dogs and a jaguar, each of which had thin tubes joining its two front feet and its two back feet. The discs lay beside them. Similar finds have been made further north, near Mexico City.

In the 1980s I saw the Tres Zapotes animals, or ones like them, at a museum in the Yucatán Peninsula. I was there with an Italian engineer whom I had met by chance a few hours before. Well before me, the engineer figured out the significance of the tubes between the figurines' feet. "Those are for axles," he said. "And those"—pointing to the discs—"must be the wheels." Looking at the little figures, it seemed obvious that they had been equipped with wheels in precisely the form he suggested.

The engineer scrunched up his face with incredulity. Tres Zapotes dated back to at least 1000 B.C. So the Olmec and their successors must have had the wheel for more than two thousand years. "Why didn't they use it for anything other than little toys?" he asked in Italian. "How could they not have understood that you could make bigger wheels and put them on carts? *Hanno fatto proprio una stupidaggine, quel tipi.*"

The word *stupidaggine* (an absurdity), similar enough in Italian and Spanish, rang out in the room, drawing stares. The engineer seemed not to care. He looked positively offended at the Olmec failure to see the world in the same way as a contemporary European engineer.

I'm giving my acquaintance a hard time, but his bafflement was easy for me to understand. In Mesopotamia, the wheel dates back to at least the time of Sumer. It was a basic part of life throughout Eurasia. Chariot wheels, water wheels, potter's wheels, millstone wheels—one can't imagine Europe or China without them. The only thing more mysterious than failing to invent the wheel would be inventing the wheel and then failing to use it. But that is exactly what the Indians did. Presumably countless thousands of people rolled the toylike figurines back and forth. How could none of them have thought of making their wheels bigger and more useful?

Some reasons are apparent. Because of the Pleistocene extinctions, the Americas lacked animals suitable for domestication into beasts of burden; without animals to haul carts, individuals on rough terrain can use skids almost as effectively. Even with animals, though, the Olmec would not have

had much use for wheeled vehicles. Their country is so wet and boggy that Stirling's horses sank to their chests in mud; boats were a primary means of transportation until recently. In addition one might note that Mesopotamian societies were not alone in their wheel-blindness. Although Mesopotamia had the wheel in about 4000 B.C., nearby Egypt did not use the wheel until two thousand years later, despite being in close contact. Still, none of this explains why no Mesoamerican society ever used wheels to make ceramics and grind maize. After all, every society in Eurasia eventually employed pottery wheels and mill wheels.

A better answer might be one implicit in Robert Temple's book, *The Genius of China*, a history of Chinese science and technology published in 1998. According to Temple, the Chinese invented the moldboard plow by the third century B.C. Made of cast iron, the plowshare was shaped like a V, with the blade carving into the ground and the two arms arcing away like gull wings. Because the arms were curved, they turned the earth away from the blade, which both reduced friction and more effectively plowed the soil. ("The 'moldboard' is the curved plowshare; the name comes from *mold*, the Old German word for soil.")

The design of the moldboard plow is so obvious that it seems incredible that Europeans never thought of it. Until the Chinese-style plow was imported in the seventeenth century, farmers in France, Germany, Italy, the Netherlands, and other states labored to shove what amounted to a narrow slab of metal through the earth. "The increased friction meant that huge multiple teams of oxen were required, whereas Chinese plows could make do with a single ox," Temple explained. The European failure to think up the moldboard, according to science historian Teresi, was "as if Henry Ford designed the car without an accelerator, and you had to put the car in neutral, brake, and go under the hood to change speed. And then we did this for 2,000 years."

European agricultural production exploded after the arrival of the moldboard plow. The prosperity this engendered was one of the cushions on which the Enlightenment floated. "So inefficient, so wasteful of effort, and so utterly exhausting" was the old plow, Temple wrote, "that this deficiency of plowing may rank as mankind's single greatest waste of time and energy." Millions of Europeans spent centuries behind the plow, staring at the blade as it ineffectively mired itself in the earth. How could none of them have thought of changing the design to make the plow more useful?

The complexity of a society's technology has little to do with its level of social complexity—something that we, in our era of rapidly changing, seemingly overwhelming technology, have trouble grasping. Every society, big or

little, misses out on "obvious" technologies. The lacunae have enormous impact on people's lives—imagine Europe with efficient plows or the Maya with iron tools—but not much effect on the scale of a civilization's endeavors, as shown by both European and Maya history. The corollary is that widespread and open trade in ideas is the best way to make up for the lacunae. Alas, Mesoamerica was limited in this respect. Like Europe, it was an extraordinarily diverse place with a shared cultural foundation. But where Europe had the profoundly different civilizations of China and Islam to steal from, Mesoamerica was alone in the world.

Or seemed to be, anyway.

A SLICE OF PERU

About a hundred miles north of the Peru-Chile border, the coastal highway passes by an uninhabited beach ringed by a tall chain-link fence. The fence has an entrance with a gigantic, stylized statue of a woman with huge earrings. By the statue hangs a faded banner: *Bolivia Mar*.

When Bolivia declared its independence it had a territorial pseudopod that extended southwest from its Andean heartland through the Atacama Desert to the sea. The land was useless for agriculture but had four plausible seaports and huge underground deposits of prehistoric guano, which Chilean companies mined and shipped to Europe for fertilizer. (Bolivia, then as now impoverished, didn't have the capital for this industry.) In 1878, Hilarión Daza, the illegitimate son of an Italian acrobat, seized power in Bolivia. Immediately he raised taxes on the Chilean-owned guano mines, which the previous Bolivian government had promised not to do. Outraged, Chile rolled its army into the area. In vain did Bolivia counterattack with its ally, Peru; Chile simply repelled their incompetently led forces and took over the entire territory, as well as a chunk of southern Peru. Ejected in an outburst of popular anger, Daza fled to Europe, taking most of Bolivia's treasury with him.

Chile finally returned most of Peru's territory in 1929 but never gave back any land to Bolivia—an outcome that nation has never accepted. To this day, Bolivia's parliament has a representative from the lost maritime province. The Miss Bolivia contest always includes a contestant ostensibly from the coast. Maps are sold in which the conquered land is still part of Bolivia.

In a gesture to its longtime, long-suffering ally, Peru symbolically gave two miles of its shoreline to Bolivia in 1992. Bolivia Mar—Bolivia-by-the-Sea—is a little island of Bolivia entirely surrounded by Peru. It has no facil-

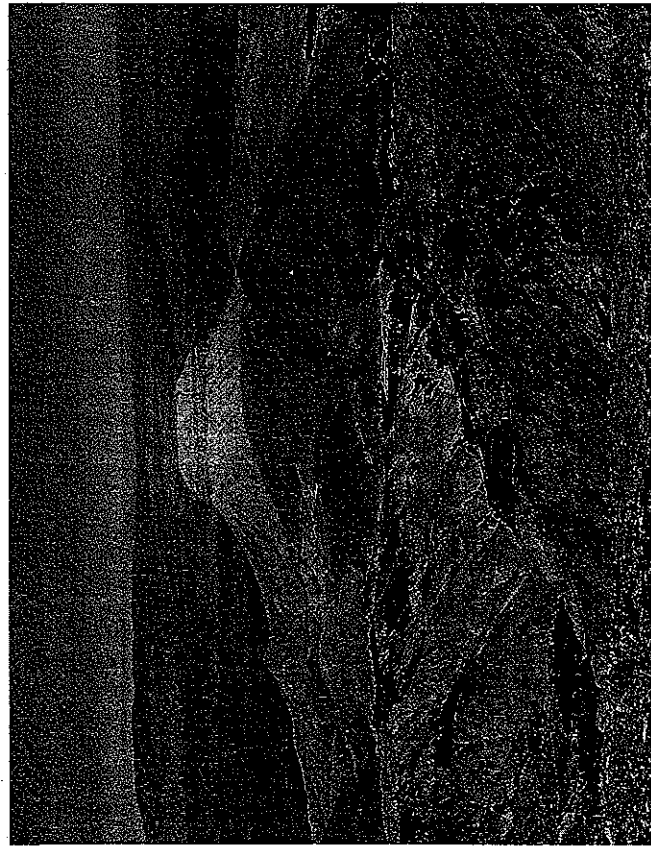
ties of any kind, so far as I could tell when I passed by. Private enterprise was supposed to build an industrial duty-free port in Bolivia Mar. Thus far the free market has not accepted the challenge. Every now and then parties of Bolivians drive down to Bolivia Mar to swim—a political gesture.

The main highway from Bolivia Mar to Bolivia itself follows the Osmore Valley, cutting a perfect sectional slice through Peru on the way. For the first fifteen miles the road climbs through a desert landscape almost devoid of settlement and prone to fog. Then the road hits a plateau and the fog dissipates. The landscape that comes into view is so dry that in most years the Osmore River simply disappears into the desert.

Around the small city of Moquegua the river hoves back into view and the highway abruptly pitches into the Cordillera Negra. The windshield fills with enough canyons, bluffs, mesas, and cliffs for a dozen *Road Runner* cartoons. Standing higher than its neighbors, at an altitude of about eight thousand feet, is a wide pillar of rock with a rounded, convoluted top that vaguely resembles the rounded, convoluted top of a human brain. The pillar is called Cerro Baúl. For about two hundred years, it was the sole meeting ground of two of the Americas' largest societies—societies similar in scale to, say, the Maya realm, but much less well known.

* The two states, Wari and Tiwanaku, were probably the greatest of the Inkas' forerunners, and certainly the predecessors from whom they took the most. In their separate ways, both were children of Norte Chico. They worshipped figures in Staff God poses, lived in networks of vertical exchange, and had public architecture with designs based on templates from the coast. But in other ways they were as different from each other as Sicily and Scandinavia. Of the two, Wari was the more conventional, centralized state. Based east of Lima in the Andes heights, it first became prominent in the sixth century A.D.—a bad time to be launching a nation on the Pacific side of South America. At about that time, Andean societies were assailed by the first of several decades-long droughts, paradoxically interrupted by El Niño-induced floods. Some polities may have disintegrated beneath the climatic assault, but Wari thrived. The principal reason for its success was its innovative techniques of terracing and irrigation, the latter being used to implement the former. Surprisingly, Peru has more arable land above nine thousand feet than below. By diverting snowmelt from the ever-present Andes icecaps to high farm terraces, Wari was able literally to rise above the drought and flooding of lower elevations.

The staple crop of the highlands was the potato, which unlike maize regularly grows at altitudes of 14,000 feet; the tubers, cultivated in hundreds of varieties, can be left in the ground for as long as a year (if the weather



An anomaly in the southern Andes foothills, the great stone of Cerro Batúl dominates the neighboring slopes. On its summit are the remains of a Wari city.

remains cold), to be dug up and cooked when needed. (The potato's cold tolerance spurred its embrace by European peasants. Not only did potatoes grow in places where other crops could not, the plant was an ally in smallholders' ceaseless struggle against the economic and political elite. A farmer's barnful of wheat, rye, or barley was a fat target for greedy landlords and marauding armies; buried in the soil, a crop of potatoes could not be easily seized.) Maize, though, was what people wanted, the grain of choice for the elite—it was what you made *chicha* from. Its prestige was another reason for Wari's success. Because terraces soak up more sunlight than steep slopes, maize can be grown at higher than usual altitudes on them; irrigation similarly increases the area available for maize farming.

In a process that Michael Moseley has likened to "patenting and marketing a major invention," the Wari passed on their reclamation techniques to their neighbors, bringing a thousand-mile-long swath of the Peruvian Andes under their cultural sway. A sign of their influence was the spread of the

Wari religion, in which the figure archaeologists call Staff God was dominant—though the Wari transformed the staff, as if to remind others of their agricultural beneficence, into a stalk of maize. By the end of the first millennium A.D., Wari techniques had reclaimed more than a million acres of cropland from mountainsides that almost anywhere else would have been regarded as impossibly dry, steep, and cold. Today three-quarters of the terraces are abandoned, and the alpine landscape has not regained the productivity it had a thousand years ago. But until the Spanish conquest Andean valleys were so thoroughly punctuated by Wari-inspired terraces that to the Jesuit Bernabé Cobo they looked "as if they were covered with flights of stairs."

Wari's capital city, also named Wari, occupied an alpine plateau near the modern city of Ayacucho. Construction began in the first few centuries A.D. The city ultimately spread across two square miles, an array of two- and three-story buildings in compounds behind massive walls. Both peasant homes and great palaces were built in similar styles, according to William H. Isbell, an anthropologist at the State University of New York at Binghamton, and Alexei Vranich, an archaeologist at the University of Pennsylvania Museum of Archaeology and Anthropology. Everything was enclosed behind high, white walls, in what the two researchers described in 2004 as a hive of "repetitive, modular cells organized in high-walled geometric blocks." There were no standout public buildings, no great public spaces, no spectacular vistas—only a thicket of walls and narrow streets strewn with garbage (archaeologists have turned up so few clean floors and surfaces, Isbell and Vranich wrote, "that it is apparent that Wari people experienced domestic refuse as benign and unthreatening"). Apparently the walls ringing and crisscrossing the city were intended for privacy, not protection; Wari was not located in an easily defended spot. Along the spine of the Andes the empire set up a string of a dozen administrative centers that were like smaller versions of the capital. These were not built with defense in mind, either. Indeed, there is little record of Wari warfare. Its supremacy was commercial and intellectual; it was based less on infantry troops than on innovative technology. All of which may explain some of its behavior in Cerro Batúl.

Wari emissaries arrived in Moquegua around 600 A.D., according to Patrick Ryan Williams and Donna V. Nash, Field Museum archaeologists who have been working there since the early 1990s. A simpler culture had already staked out the best farmland in the area. The Wari neither aggressively threw them out nor withdrew in dismay. Guided, one imagines, by instructions from headquarters, they quickly set up living quarters on Cerro

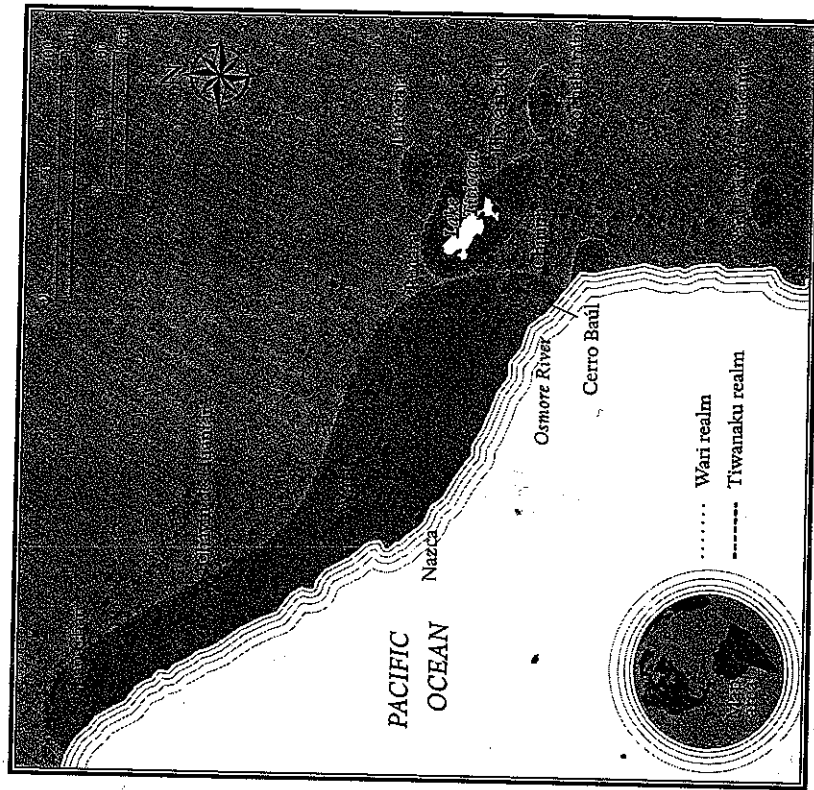
Baúl itself. The big mesa is to this day regarded as an *apu*, an ancient spirit transfigured into rock. Thus putting a city directly on top of it was an arresting statement: *Here we are*.

On a practical level, living on a five-hundred-yard-long mesa was a daunting task. To supply water, the Wari carved a fifteen-mile canal through the mountains from the peaks to the bottom of Cerro Baúl, an engineering feat that would be a challenge today. "And even that got water only to the bottom of the hill," Williams told me. "After that, it was bucket brigades." As I huffed and puffed on the wickedly steep, thirty-minute hike to the summit, he invited me to imagine a continuous line of servants exchanging ceramic jugs (slopping, brimful ones going up; light, empty ones going down) along the path, working day in and day out to provide water for the priests and princes above.

Small, rudely fashioned models of farmhouses and farmyards covered the top of the butte. Most of the models simply outlined walls, fences, and doors with loose stones, but some were elaborate constructions complete with plastic model cars, toy animals, and thatch roofs. People were climbing Cerro Baúl, building their maquettes, and praying that the heavens would give them the real-life equivalents. The miniature farms extended for hundreds of yards in all directions. Here and there, makeshift crosses and pictures of saints added a veneer of Catholicism to indigenous Andean belief. Some of the ruined Wari walls were covered with ruined model walls. "This is getting out of hand," Williams said. "I don't want to knock down somebody's dream house to get to an archaeological site."

In about 750 A.D., about a century after Wari came to Cerro Baúl, Tiwanaku groups infiltrated the region around it. In most places on earth, this encounter would have been fraught with tension. And perhaps if Tiwanaku had been more like Wari there would have been immediate war. But Tiwanaku was so different in so many ways that ordinary expectations rarely apply to it. The celebrated anthropologist Clifford Geertz has half-jokingly suggested that all states can be parceled into four types: pluralist, in which the state is seen by its people as having moral legitimacy; populist, in which government is viewed as an expression of the people's will; "great beast," in which the rulers' power depends on using force to keep the populace cowed; and "great fraud," in which the elite uses smoke and mirrors to convince the people of its inherent authority. Every state is a mix of all of these elements, but in Tiwanaku, the proportion of "great fraud" may have been especially high. Nonetheless, Tiwanaku endured for many centuries.

Tiwanaku's capital, Tiwanaku city, was at the southwest end of Lake Titicaca. Situated at 12,600 feet, it was the highest city in the ancient world.



WARI AND TIWANAKU, 700 A.D.

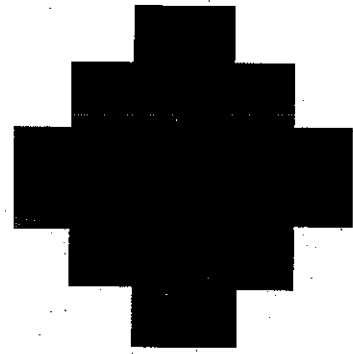
Today visitors from lower altitudes are constantly warned that the area is very cold. "Bring warm clothing," Williams advised me in Cerro Baúl. "You're going to freeze." The warnings puzzled me, because Lake Titicaca, which is big enough to stay at a near-constant 51 degrees Fahrenheit, moderates the local climate (this is one reason why agriculture is possible at such height). On winter nights the average temperature is a degree below freezing—cold, but not any colder than New England, and considerably warmer than one would expect at 12,600 feet. Only when I traveled there did I realize that this was the temperature *indoors*. The modern town of Tiwanaku is a poor place and few of its buildings have any heating. One night there I attended a performance at a family circus that was touring the Andes. It was so cold inside the tent that for the first few minutes the audience was

shrouded in a cloud of its own breath. My host that night was an American archaeologist. When I woke the next morning in her spare bedroom, my host, in parka, hat, and gloves, was melting water on her stove. The cold does not detract from the area's beauty: Tiwanaku sits in the middle of a plain ringed by ice-capped mountains. From the ruin's taller buildings the great lake, almost ten miles to the northwest, is just visible. The wide expanse of water seems to merge into the sky without a welt.

The first important settlement around Titicaca was likely Chiripa, on a little peninsula on the lake's southwest coast. Its ceremonial center, which may date to 900 B.C., was built around a Norte Chico-style sunken plaza. Chiripa was one of half a dozen small, competitive centers that emerged around the lake in that time. Most depended on raised-field agriculture, in which farmers grow crops on flat, artificially constructed surfaces created for the same reason that home gardeners grow vegetables on raised beds. (Similar but even larger expanses of raised fields are found in the Beni, the Mexican basin, and many other places.) By the time of Christ's birth, two of these early polities had become dominant: Pukara on the northern, Peruvian edge of the lake and Tiwanaku on the opposite, Bolivian side. In the third century A.D., Pukara rather abruptly disintegrated politically. People still lived there, but the towns dispersed into the countryside; pottery making, stela carving, and monument building ceased. No one is certain why.

Although Tiwanaku has been occupied since at least 800 B.C., it did not become an important center until about 300 B.C. and did not expand out from southern Titicaca until about two hundred years after Pukara's decline. But when it did reach out to its south and west, Tiwanaku transformed itself to become what Alan Kolata, an archaeologist at the University of Chicago, has called a "predatory state." It was not the centrally administered military power that term conjures. Instead, it was an archipelago of cities that acknowledged Tiwanaku's religious preeminence. "State religion and imperial ideology," Kolata argued, "performed much the same work as military conquest, but at significantly lower cost." Awed by its magnificence, fearful of the supernatural powers controlled by its priesthood, local rulers subordinated themselves.

Central to this strategy of intimidation was Tiwanaku city. A past wonder of the world, it is badly damaged today. In the last two centuries, people have literally carted away many of its buildings, using the stones for churches, homes, bridges, public buildings, and even landfill. At one point the Bolivian government drove a railroad through the site (recently it laid a road through another part). Hands more enthusiastic than knowledgeable reconstructed many of the remaining buildings. Still, enough remains to get a sense of the ancient city.



The Andean cross

Dominating its skyline was a seven-tiered pyramid, Akapana, laid out in a pattern perhaps inspired by the Andean cross. Ubiquitous in highlands art, the Andean cross is a stepped shape that some claim is inspired by the Southern Cross constellation and others believe represents the four quarters of the world. Whatever the case, Akapana's builders planned with a sense of drama. They constructed its base walls with sandstone blocks, the array interrupted every ten feet by rectangular stone pillars that are easily ten feet tall. So massive are the pillars that the first European to see Tiwanaku, Pedro Cieza de León, later confessed himself unable to "understand or fathom what kind of instruments or tools were used to work them." Rising from the center of a large moat, Akapana mimicked the surrounding mountains. A precisely engineered drainage system added to the similarity by channeling water from a cistern-like well at the summit down and along the sides, a stylized version of rainwater plashing down the Andes.

Atop an adjacent, somewhat smaller structure, a large, walled enclosure called Kalasasaya, is the so-called Gateway of the Sun, cut from a single block of stone (now broken in two and reassembled). Covered with a fastidiously elaborate frieze, the twelve-foot gateway focuses the visitor's eye on the image of a single deity whose figure projects from the lintel: the Staff God.

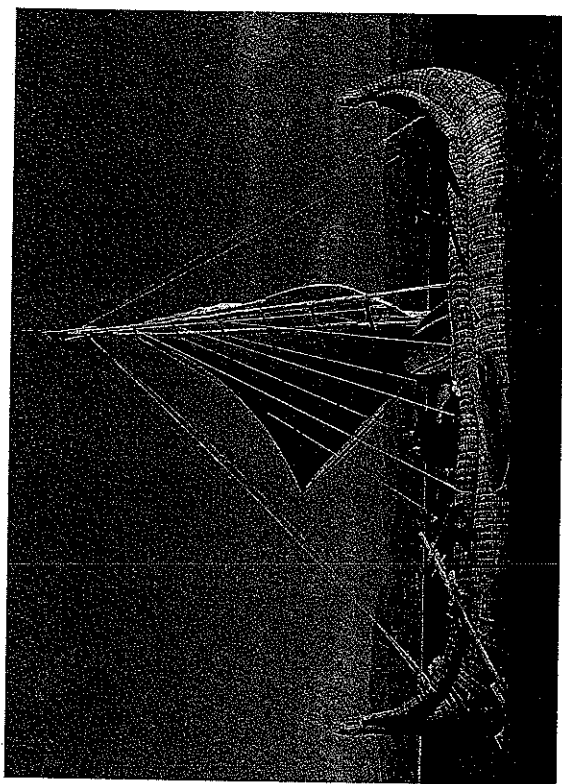
Today the Gateway of the Sun is the postcard emblem of Tiwanaku. During the winter solstice (June, in South America) hundreds of camera-toting European and American tourists wait on Kalasasaya through the entire freezing night for the sunrise, which is supposed to shine through the Gateway on that date alone. Guides in traditional costume explain that the reliefs on the lintel form an intricate astronomical calendar that may have been brought to earth by alien beings. To keep themselves warm during the inevitable longueurs, visitors sing songs of peace and harmony in several languages. Invariably the spectators are stunned when the first light of sunrise appears well to the side of the Gateway. Only afterward do they discover that the portal is not in its original location, and may have had nothing to do with astronomy or calendars.

If those tourists had come to Tiwanaku at its height, walking through the miles of raised fields surrounding it to the city's carefully fitted stone walls,

they would have been delighted by its splendor. But it might also have seemed curiously incomplete, with half the city falling down and in need of repairs and the other half under construction. Modern drawings of ancient cities tend to show them at an imagined apogee, the great monuments all splendidly arrayed together, perfect as architectural models. But this is not what Tiwanaku looked like, nor even what it was *meant* to look like, according to Isbell and Vranich. From the very beginning, the two men wrote in 2004, the city was partly in ruins—intentionally so, because the fallen walls bequeathed on Tiwanaku the authority of the past. Meanwhile, other parts of the city were constantly enveloped in construction projects, which testified to the continued wealth and vitality of the state. Sometimes these projects acquired construction materials by cannibalizing old monuments, thereby hastening the process of creating ruins. In the Andean tradition, labor was probably contributed by visiting work parties. Periodically ritual feasts that included much smashing of pottery interrupted the hubbub of construction. But it always continued. “They build their monuments as if their intent was never to finish them,” the Spanish academic Polo de Ondegardo marveled in 1971. Exactly right, Isbell and Vranich said. Completion was not the object. The goal was a constant buzz of purposeful activity.

As our hypothetical modern visitors wandered through the hurly-burly of construction and deconstruction, they might have felt that despite the commotion something was missing. Unlike Western cities, Tiwanaku had no *markets*—no bazaars full of shouting, bargaining, conniving entrepreneurs; no street displays of produce, pottery, and plunk; no jugglers and mimes trying to attract crowds; no pickpockets. In Africa, Asia, and Europe, Kolata wrote, “a city was a place of meeting and of melding for many different kinds of people. . . . Through trade and exchange, through buying and selling of every conceivable kind, the city was made and remade.” Tiwanaku was utterly different. Andean societies were based on the widespread exchange of goods and services, but kin and government, not market forces, directed the flow. The citizenry grew its own food and made its own clothes, or obtained them through their lineages, or picked them up in government warehouses. And the city, as Kolata put it, was a place for “symbolically concentrating the political and religious authority of the elite.” Other Andean cities, Wari among them, shared this quality. But Tiwanaku carried it to an extreme.

Tiwanaku has been excavated for a century, and the more archaeologists delve into it the less there seems to be. To Vranich, the capital’s lack of resemblance to European imperial cities extends well beyond the absence of marketplaces. Far from being the powerful administrative center envisioned by earlier researchers, he says, Tiwanaku was a combination of the Vatican



On Lake Titicaca, the reed boats known as *totora* are still in use, as they have been for two thousand years. This replica of a large *totora* was built in 2001 to prove the vessels could have hauled the big stones used in Tiwanaku’s walls.



The so-called Gateway of the Sun attracts pilgrims by the thousands who seek astronomical meaning in its location. Unfortunately, it was moved to its present site in the twentieth century.

and Disneyland, a religious show capital with a relatively small population—almost a *staff*—that attracted pilgrims by the thousand. Like the tourists at the solstice today, visitors came to this empire of appearances to be dazzled and awed. "In the central city, buildings and monuments went up and down, up and down, at an incredible rate," Vranich told me at Tiwanaku, where he had been working since 1996. "Nothing ever got finished completely, because they were just concerned with the facades. They had to keep changing the exhibits to keep the crowds coming."

The encounter between Tiwanaku and Wari at Cerro Baúl seems to have gone remarkably smoothly. At any rate, a study of more than a thousand Wari and Tiwanaku graves found no evidence of the trauma associated with violence. Instead, the two societies split the region between them. Wari camped atop Cerro Baúl and a neighboring hillock, Cerro Mejía. Between them was a steep valley with Tiwanaku settlements scattered throughout. Because Wari and Tiwanaku pottery differed, Williams and Nash have been able to map which group lived in which neighborhood by the distribution of ceramic fragments. The Wari canal provided drinking water, but had to pass through Tiwanaku territory at the base of Cerro Baúl. Tiwanaku let the water through, but took enough to irrigate more than seven hundred acres of terraces.

At the same time, Wari and Tiwanaku kept themselves separate. Although they shared resources, there is little evidence that people from one culture visited the other often, or had friendships across the political lines. Wari homes were furnished with Wari goods; Tiwanaku homes, Tiwanaku goods. Despite living next to each other, people continued to speak their different languages and wear their different clothing and look for inspiration and instruction from their different capitals. The social-science word for such intermingling without intermixing is "interdigitization." For two centuries at Cerro Baúl, Wari and Tiwanaku were like people in parallel worlds, sharing the same time and space but implacably separate from each other. It is a small reminder that Indians were neither the peaceful, love-thy-neighbor types envisioned by some apologists or the brutal, ceaselessly aggressive warriors decried by some political critics.

Wari abandoned Cerro Baúl in about 800 A.D., ceremoniously burning its buildings and smashing pottery. The retreat was part of a general fall, soon followed by Tiwanaku. Both declines have been laid to drought, but this is contested. The Wari had already survived drought. As for Tiwanaku, Vranich said, "How much would drought matter to Disneyland?" Its ability to retain its audience would be far more important.

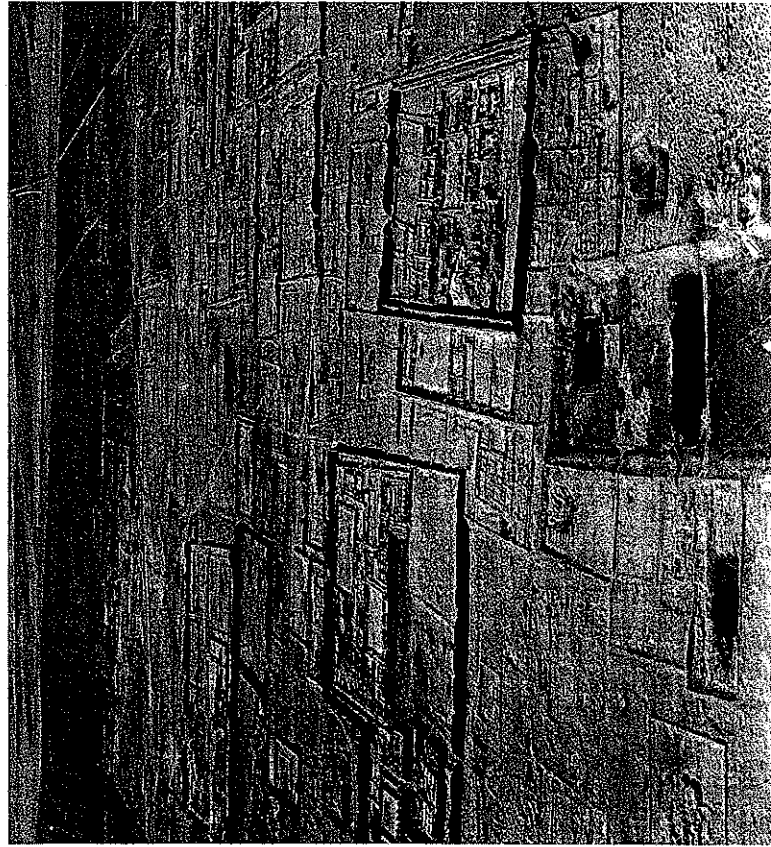
The successors to both Wari and Tiwanaku combined the former's orga-

nizational skills and the latter's sense of design and razzle-dazzle. First came Chimor, then the greatest empire ever seen in Peru. Spread at its greatest extent over seven hundred miles of the coastline, Chimor was an ambitious state that grew maize and cotton by irrigating almost fifty thousand acres around the Moche River (all of modern Peru only reached that figure in 1960). A destructive El Niño episode in about 1100 A.D. made irrigation impossible for a while. In response, the government forced gangs of captive laborers to build a fifty-three-mile, masonry-lined canal to channel water from the Chicama River, in the next valley to the north, to farmland in the Moche Valley. The canal was a flop: some parts ran uphill, apparently because of incompetent engineering, and the rest lost nine-tenths of its water to evaporation and seepage. Some archaeologists believe that the canal was never meant to function. It was a PR exercise, they say, a Potemkin demonstration by the Chimor government that it was actively fighting El Niño.

When the bad weather ended, Chimor looked outside its borders. Armies went out and returned victorious to Chan Chan, the Chimor capital, a seaside metropolis with a dense center that covered four square miles. Dominated by nine high-walled imperial palace-tombs and five cathedral-like ceremonial complexes, the city was both exemplary in its grandeur and oddly empty, because its streets were restricted to the elite. Commoners were barred, except for a few specialized technicians and craftworkers. Each palace was hundreds of feet on a side and many were three stories tall. They were filled with storage space—living quarters were almost an afterthought. Their great beams adorned with splendidly worked gold and silver, the huge structures were jammed together around the center of town like people huddling in the shelter of an awning.

Chan Chan suffered a palace surfeit because dead rulers were regarded as divine figures. As with the Inka, the kings' mummified bodies continued to live opulently in their own homes and could not be displaced; indeed, the mummies were necessary presences at important state occasions. As a result, each new ruler had to build his own palace and acquire the riches necessary to maintain it till the end of time. The system almost guaranteed imperial ambitions and exuberant construction plans.

The biggest of Chan Chan's surviving palaces may have belonged to Minchacaman—the eleventh king in the Chimor dynasty, according to one Spanish account—who reputedly conquered much of the coastline. Minchacaman was a powerful figure who could have taken over even more land than he did. Unfortunately for him, he lived at the same time that a previously little-known group, the Inka, acquired a new ruler, Pachakuti. In about 1450 the Inka army, led by Qhapaq Yupanki, Pachakuti's brother, besieged the



In this rare aerial photograph—taken in 1937, before modern looting blasted the site—the ruined Chimor capital of Chan Chan sprawls across the northern Peruvian coast. One of the wonders of the fifteenth-century world, Chan Chan abruptly fell to the Inka in about 1450; eighty years later, Spanish diseases and Spanish soldiers destroyed much of what had survived the Inka.

city-state of Cajamarca, in the foothills east of Chimor. Cajamarca's leader had allied himself with Minchacaman, who rushed to his aid with an army. He does not seem to have known what he was in for, possibly because he viewed the Inka as a gang of rustic thugs. Qhapaq Yupanki awaited him in an ambush. Minchacaman and his army were forced to flee as Cajamarca fell to the Inka. Qhapaq Yupanki covered himself with so much glory that when he returned home to Qosqo his brother, sensing future trouble, promptly executed him.

A decade or so later—in 1463, if Spanish chronicles are correct—the emperor sent out another army led by his son and designated successor,

Thupa Inka Yupanki. By that time nobody thought of the Inkas as hicks. Thupa Inka descended the Moche River and paralyzed Chimor's defenses by the simple expedient of threatening to destroy its water supply. Minchacaman was captured, taken to Qosqo, and forced to watch Thupa Inka's victory celebration. Chimor's conquerors were quick studies. Liking the courtly magnificence of Chan Chan, they hauled away what they could and, more important, forced the city's gold, silver, and gem workers to accompany them to Qosqo. They were instructed to transform the city into a new Chan Chan, only more impressive. Seven decades later, when Pizarro held his victory celebration in Qosqo, it was more lushly opulent than any city in Europe.

MAKING THE WAK'A

Jonathan Haas and Winifred Creamer took me to Caballote, a narrow, dusty bowl, perhaps half a mile long and a quarter mile wide, a few miles up the Fortaleza River from the Peruvian coast. At the mouth of the bowl were three mounds in a rough semicircle that faced a fourth mound. In front of one mound, like a one-eighth scale model of Stonehenge, was a ragged circle of *wak'a*: sacred stones. Not far from the *wak'a* looters had dug up an ancient graveyard, pulling out the bodies and unwinding them from their sheets in a search for gold and jewels. When they didn't find any, they threw the bones down in disgust. In a square perhaps fifty yards on a side the ground was carpeted with broken human bones and scraps of thousand-year-old fabric.

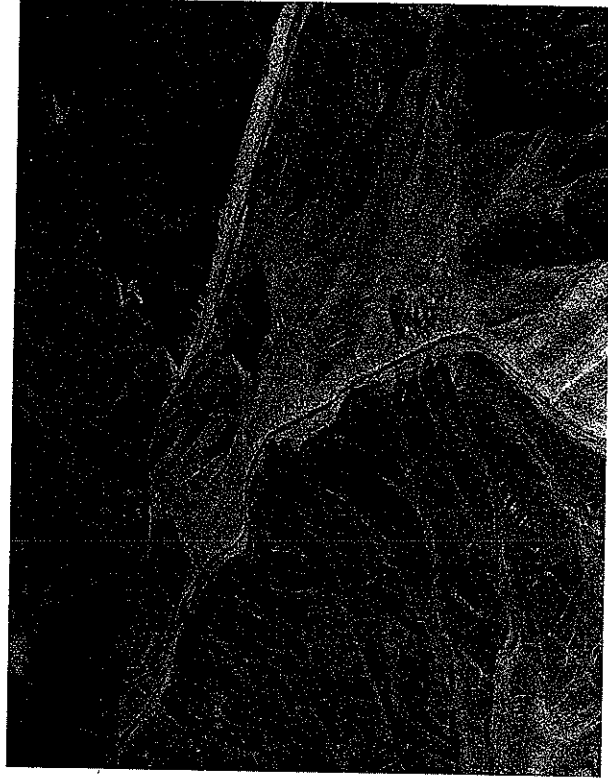
We walked a little further and were greeted by a curious sight: skulls from the cemetery gathered into several small piles. Around them were beer cans, cigarette butts, patent-medicine bottles, half-burned photographs, and candles shaped like naked women. These last had voodoo pins stuck in their heads and vaginas. Local people came to these places at night and either dug for treasure or practiced witchcraft, Haas said. In the harsh afternoon light they seemed to me tacky and sad. I imagined that the families of the people who had been buried at Caballote so long ago would have been outraged if they could have known what would happen to the bodies of their loved ones.

But then it occurred to me that my views may not have been shared by either the present or the past inhabitants of Norte Chico. I had no idea what people in Wari or Chimor would have thought of the scene before me. So far in this section, I have mainly described the economic and political history of Andean Indians. But people live also in the realm of the affective and aesthetic—that's why they bury bodies and sometimes dig them up and pour

love potions on them. Despite all the knowledge gained by scientists in the last few decades, this emotional realm remains much harder to reach.

An obvious example on the southern coast is the Nazca, famous for the huge patterns they set into the ground. Figures of animals and plants, almost a thousand geometric symbols, arrow-straight lines many miles long—what were they for? Peruvian anthropologist Toribio Mejía Xesspe first brought these famous drawings to the attention of the outside world in 1927. Four decades later, the Swiss writer Erich von Däniken set off an international furor by claiming that the Nazca Indians could not have made these symbols, because they were too big for such “primitive” people to construct, and because they are visible only from the air. Instead, he said, the giant figures were landing signals for space travelers; the whole plain was a sort of gigantic extraterrestrial airport. Expanded in a series of best-selling books, this notion turned the lines into a major tourist attraction. Exasperated scientists pointed out that a) small groups could have constructed the images by moving the dark surface stone to expose the lighter-colored earth beneath, and b) the Nazca did not have to see the figures to experience them, for they can be understood by walking the lines, which it is believed the Indians did. The prevailing theory today is that the straight lines mapped out the area’s many underground faults, which channel water. But nobody knows why the Nazca made the animal and plant figures, which seem less likely to have a direct function. What were the Nazca thinking as they created them? How did they feel when they walked them? To this day, the answers remain frustratingly far away.

Or consider the Moche, leaders of a military state that overran much of the northern coast, submerging the identities of its victims in its own. Huaca del Sol, the Moche capital, contains the largest adobe structure in the Andes, still hauntingly evocative despite centuries of systematic looting. (Unwilling to laboriously dig their way through the palace’s tombs, the Spaniards diverted the Moche River through it, washing out the riches in orthodox Augean-stable style; contemporary thieves have contented themselves with picks and shovels.) After about 300 A.D., Moche artists confined themselves to perhaps half a dozen subjects, painting stories of supernatural figures on pottery and murals with ever more naturalistic technique. Actors reenacted the same stories in grand pageants and ritual celebrations. Individual combat is a common theme; losers were formally stripped of their garments and forced to parade naked. Another oft-repeated tale involves the death and burial of a regal figure. Many of the people in the paintings are sharply individualized. Great effort has gone into studying the Moche, but as Moseley says, their identities and motives often remain “elusive.” The Moche polity broke up around 800 A.D., taking with it our chance to understand.



The history of Andean societies is so rich and complex that it often leaves archaeologists feeling overwhelmed—there is so much to learn that they can never keep up. A single example: scientists did not confirm the existence of the Great Wall of Peru, a forty-mile stone rampart across the Andes, until the 1930s. And it still has never been fully excavated.

One of the few moments when I imagined I could encompass something of the inner lives of these long-ago people occurred in Chavín de Huantar, a city of several thousand people that existed between about 800 B.C. and 200 A.D. Its most important feature, a ceremonial temple shaped in a Norte Chico-style U, was a masterpiece of architectural intimidation. Using a network of concealed vents and channels, priests piped loud, roaring sounds at those who entered the temple. Visitors walked up three flights of stairs, growls echoing around them, and into a long, windowless passageway. At the end of the corridor, in a cross-shaped room that flickered with torchlight, was a fifteen-foot-high stone figure with a catlike face, taloned fingers, fierce tusks, and Medusa hair. Nobody today is sure of the god's identity. Immediately above it, hidden from visitors' eyes, sat a priestly functionary, who provided the god's voice. After the long, torchlit approach, walking straight into the gaze of the snarling deity, mysterious bellows reverberating off the stone, the oracular declamation from above must have been spine-chilling.

Most of the complex is open to tourists. Many of the sculptures have

been put into museums; others presumably have been looted. Yet walking into the temple still felt to me like entering a mountain of solid rock. Over and over again, Andean stories tell of spirits embodied in stones and giants transformed into natural features. The landscape has an intricate numinous geography; it is charged with meaning that must be respected and heeded. The earth, in this view, is not something to be left alone; the *wak'a* that litter Peruvian anthropological sites are often partly sculpted, as if they had needed some human attention to manifest their sacred qualities. Thus the human-made tunnels into the temple were part of what made it embody the power of a mountain. As I walked down the dimly lighted corridor toward where the torchlit deity had stood, my fingers ran along the walls created by Chavín craftworkers. They were fit beautifully into place and as cold and hard as the mountains they came from. But they did not gain their power without my hand to close the circuit. The natural world is incomplete without the human touch.

PART THREE

Landscape with Figures



back the date to about 4200 B.C. (Piperno and Flannery 2001; Benz 2001). Both of these sites contained fully domesticated maize (Benz and Illis 1990). Pope et al. found teosinte pollen grains as early as 5100 B.C. in a wet Gulf Coast site where teosinte is not native, suggesting that it was moved there from the highlands. By 4000 B.C. the pollen is dominated by modern maize (Eubanks 2001a; MacNeish and Eubanks 2000; Pope et al. 2001).

196 "arguably man's"; Pederoff 2003.

196 Diversity of maize: Doebley, Goodman, and Stuber 1998. The reason for the diversity is that the ancestor species were hyperdiverse (Byrne-Walker et al. 1998).

197 Aragón Cuevas research: Author's interviews, Aragón Cuevas. The number of landraces varies from study to study, because the term is not precisely defined. It is often claimed that more than two hundred exist in Latin America (e.g., Wellhausen et al. 1957, 1952).

197 Five thousand cultivars: Author's interview, Wilkes. This is a widely cited guess by a distinguished researcher with long experience in the field.

197 Chamula statistics: Anon. ed. 1998a. The data are from 1991, the most recent year for which census results are available.

197 Perales's study: Author's interview, Perales.

197 Milpa: Here I describe the Mesoamerican variant of an ideal. Milpa-style agriculture occurs in much of South America, though centered often on potatoes or manioc instead of maize. Even in Mesoamerica, plenty of actual milpas are nothing more than maize fields, especially where farmers grow maize for the market. Subsistence-farm milpas I have seen tend to be more diverse. Milpa cultivation is often described—incorrectly, according to Wilkes (author's interview)—as synonymous with "slash-and-burn," in which farmers clear small areas for short times and then let them go fallow (e.g., the otherwise useful Ewell and Sands 1987). Slash-and-burn, though, is generally a modern innovation (see chap. 9). A good description of the milpa is Wilken 1987. A classic early study is Cook 1921.

198 Green Revolution and milpa: Author's interviews, Denevan, Hallberg, Perales, Wilkes ("most successful"), James Boyce; Mann 2004.

199 Abundance of wild wheat and barley: Harlan and Zohary 1966 ("square kilometers," "Over many thousands," 1078).

200 "the key": Coe 1968:26.

200 Kirkby's estimate: Kirkby 1973.

200 Maize iconography: Fields 1994.

201 Maize in Europe: Crosby 2003b:180–81; Warman 2003: 97–111.

201 Pellagra in Europe, Goethe: Roe 1973; McCollum 1957:302; Goethe 1962:33–34.

201 Maize and slavery: Author's interviews, Crosby; Crosby 2003b:186–88; 1994:24.

202 Oaxaca data: Anon. ed. 1998b:332–68. The data are from 1991, the most recent year for which census results are available.

202 Estimated productivity of Green Revolution maize in Oaxaca: Author's interviews, Aragón Cuevas, James Boyce. The estimate is roughly confirmed by the calculations of Ackerman et al. (2002:36) that "a 1 percent increase in use of improved varieties was typically associated with an increase in yield of 0.037 tons/ha" and hence a 100 percent switchover is a jump of 3.7 tons/ha.

202 Economic problems of landrace maize in Oaxaca: Author's interviews, Aragón Cuevas, Bellon, Boyce, Hallberg, Ramírez Leyva, Wilkes.

7 / Writing, Wheels, and Bucket Brigades

204 Stirling's find of dated stela: Stirling 1999 ("knees," "hurried," 213), 1940a; Coe 1976b. Stirling's position was held previously by William Henry Holmes, scourge of amateur "relic

hunters." In his *National Geographic* article, Stirling says the first giant head was discovered in 1858; others put the find at 1862 (Bernal 1969:29). Following an earlier, mistaken understanding of the Mesoamerican calendar, Stirling believed that the stela was earlier than now thought; I use the modern date. Because carbon-dating had not yet been invented, he had the dates of Maya emergence wrong, too. Still, he was right to be puzzled by the Olmec.

205 Second Veracruz trip, first Olmec article: Stirling 1940b ("designed," "The ticks," 312; "basic civilization," 333; "mysterious," 334).

207 The Olmec: Among the few general book-length overviews are Coe 1996 (especially valuable for its illustrations of Olmec art); Pina Chan 1989; and Bernal 1969 (1968), the last still surprisingly useful despite its age. All espouse the "mother culture" view, which has come under increasing fire.

207 "enigmatic people": Baird and Balthous 2004:727. Similar language can be found in the *Eyewitness Travel Guide: Mexico* (New York: Dorling Kindersley, 2003), 254. These characterizations are common in the popular press (e.g., Stuart 1993a ["the Olmec stand for many as a kind of 'mother culture' to all the civilizations that came after, including the Maya and the Aztec," 93]; Lemonick 1996 ["More than 1,500 years before the Maya . . . the mysterious Olmec people were building the first great culture of Mesoamerica," 56]). Olmec emerge abruptly: Some researchers have hypothesized that Olmec society was stimulated into existence by a migration from the Pacific coast, but recent ceramics research in Veracruz casts doubt on this idea (Arnold 2003).

207 "quantum change": Meggers 1975:17.

207 "There is now little doubt": Coe 1994:62.

207 Bad name: Bernal 1969:11–12. Actually, the name is even worse than I indicated. "Olmec" doesn't refer to a people, but to the political phenomenon that began and ended with their cities. The people in those cities may still be around, but called something else.

207 Mixe-Zoquean: Campbell and Kaufman 1976.

208 Olmec rubber: Hosler, Burkett, and Tarkanian 1999; Rodríguez and Ortiz 1994.

208 1800 B.C.: Rust and Sharer 1988.

208 San Lorenzo: Coe and Diehl 1980; Cyphers ed. 1997.

208 Olmec theology: Rely 1994.

209 Thrones changed into sculptures: Porter 1989.

209 Africa-Olmec and Shang-Olmec connection: For the Africa-Olmec connection, see Barton 2001; Winters 1979; Van Sertima 1976. Van Sertima's claims are attacked in Haslip-Viera, Ortiz de Montellano, and Barbour 1997. For the Shang-Olmec connection, see Xu 1996; Meggers 1975; Ekholm 1969. I am grateful to Mike Xu for sending me a copy of his manuscript. Meggers was critiqued in Grove 1977 and responded in Meggers 1977.

209 Olmec sculptures of fetuses and pathological conditions: Tate and Bendersky 1999; Dávalos Hurtado and Ortiz de Zárate 1993. I am grateful to the Bancroft Library staffers who went to considerable trouble to find the second article for me.

209 Olmec appearance: Bernal 1969:76–79.

210 Mirrors: Heizer and Gullberg 1981.

210 Destruction of sculptures: Grove 1981.

210 La Venta: Rust and Sharer 1988. A succinct description is in Bernal 1969:35–43.

211 "not only engendered," "established the pattern": Bernal 1969:188. A well-argued contemporary version of this view is Diehl 2005.

211 Competitive interaction in Mesoamerica: Flannery and Marcus 2000 ("chiefdoms in the Basin," 33). I thank Joyce Marcus for walking me through these ideas.

211 Zapotec rise: Blanton et al. 1999; Marcus and Flannery 1996; Flannery and Marcus 2003 ("virtually unoccupied," 11802; radiocarbon dates, 11804); Spencer 2003.

- 213 Oldest writing: Marcus pers. comm. (750 B.C.). 1976 (glyphs and translation); Flannery and Marcus 2003. See also, Serrano 2002. The reason I have called the temple carving the first "securely" dated writing is that two other candidates for the title of first written text exist, but neither can be dated accurately because their archaeological context is unknown. Both are from the Tlatilco culture north of present-day Mexico City; they may have been made as early as 1000 B.C. One seal shows three glyphs that some think resemble Olmec writing. The other, a true mystery, bears what look like letters in a script of which there are no other extant examples (Kelley 1966). A third candidate for earliest Olmec writing exists. In the 1990s a team led by Mary Pohl of the University of Florida discovered a cylindrical greenstone seal two miles from La Venta. Dated to 650 B.C., the seal bears a bas-relief bird with a comic-book speech bubble bursting from its mouth. Pohl and two colleagues identified the glyphs in the bubble as precursors to the Mayan glyphs for the date 3-Ajaw (Pohl, Pope, and von Nagy 2002; Stokstad 2002). The identification is controversial. The text cannot be Mayan, they say, because Maya civilization was not firmly set in place until centuries later. Nor can it be Olmec, because other Olmec texts seem not to be related to Maya glyphs. According to John Justeson, a linguistic anthropologist at the State University of New York in Albany who has deciphered other Olmec texts, "Although many accept the greenstone glyphs as plausibly being writing, what [Pohl's team] read as a ritual calendar date on the ceramic is scarcely accepted by anyone" (email to author).
- 214 Inanna temple example: Urton 2003:15-16.
- 215 Zero as number: Teresi 2002:79-87 (GPA example, 80).
- 216 Tres Zapotes date: In fact, the initial 7 (the *bakm* figure) was missing, because the stela was broken. Stirling guessed that it was a 7, a supposition that was proven correct in 1972, when the other part of the stela was discovered (Cohn 1972).
- 217 Tentative assignment: In *The Olmec World*, for instance, Bernal never directly says that the Olmec invented zero. He merely describes the Long Count, remarking that it "necessarily implies knowledge of the zero" (Bernal 1969:114).
- 218 More than a dozen systems of writing: Coe 1976a:110ff. Coe lists thirteen forms, but does not include Olmec and whatever is on the Tlatilco seals.
- 219 Deciphered Olmec stela: Stuart 1993a, 1993b; Justeson and Kaufman 1993; 1997; 2001 (Chiapas potsherd translation, 286).
- 218 Monte Alban dispute: I have borrowed the formulation in Zeitlin 1990. Some argue that not enough data exist to resolve the question (O'Brien and Lewarch 1992).
- 218 Slabs as slain enemies: Marcus 1983:106-08, 335-60.
- 218 Fight with Tilcayote: Spencer and Redmond 2001.
- 219 Nudzahui marriage politics: Spores 1974.
- 219 8-Deer's story: Pohl 2002; Byland and Pohl 1994:119-60, 241-44; Caso 1977-79 (vol. 1):69-83, (vol. 2):169-84; Smith 1962; Clark 1912.
- 222 Wheeled toys: Stirling 1940b:310-11, 314; Charnay 1967:178-86.
- 223 Egypt and wheel: Wright 2005:46.
- 223 Moldboard plow: Temple 1998 ("so inefficient," 16). I am grateful to Dick Teresi for directing me to this book and example ("as if Henry Ford"; e-mail, Teresi to author).
- 223 Lack of relation between technology and social complexity: I lift this point bodily from Webster 2002:77. See also Ildre 2000.
- 225 Osmore Valley geography: I am grateful to Mike Moseley and Susan DeFrance for guiding me through this area, and to Patrick Ryan Williams and Donna Nash for showing me Moquegua and Cerro Baúl.
- 225 Wari and Tiwanaku: A succinct overview is La Lone 2000. Surprisingly little has been written about Wari. Among the few recent books are Isbell and McEwan eds. 1991 and Schreiber 1992. The most widely cited recent works on Tiwanaku are Kolata 1993 and

- Kolata ed. 1996-2003. William Isbell has pointed out that the two names refer simultaneously to cities, states, and religions. He has suggested that the Spanish names *Tiahuanaco* and *Huari* be used for the physical ruins and the Aymara and Ruma Suni spelling *Tiwanaku* and *Wari* be used for these polities' political and cultural styles (Isbell 2001:457).
- 225 Sixth-century climatic disaster: Fagan 1999:Chap. 7. The major apparent victims were the Moche, who flourished in a three-hundred-mile strip along the northern coast after A.D. Drought put Moche society in crisis; El Niño rains led to floods that destroyed entire villages and canal systems. El Niño also changed ocean current patterns to deposit river sediments on the shore. These quickly turned to dunes, which winds blew toward the Andes, threatening farmland. The Moche tried to regroup—and failed.
- 225 Potato's advantages and status vis-à-vis maize: McNeill 1991; Murra 1960.
- 227 Wari religion: It should be emphasized that the common image of the Staff God did not mean that the deity meant the same thing in every culture (Makowski 2001).
- 227 Terracing, arable land, abandonment: Peruvian ecologist Luis Masson has estimated that 1.2 to 1.4 million acres was terraced on just the west side of the Andes, 75 percent of which is now abandoned (pers. comm., cited in Denevan 2001:173-75); Cobo 1990:21 ("flights of stairs"); Moseley 2001:230-38 ("patenting and marketing," 233; prospering on Wari despite climatic assault, 232).
- 227 Isbell-Vranich article: Isbell and Vranich 2004 ("repetitive," 170).
- 228 Wari and Tiwanaku in Cerro Baúl: Interviews, DeFrance, Moseley, Nash, Williams, Williams, Isla, and Nash 2001.
- 229 Geertz's four states: Geertz 1980:121-22.
- 230 Chiripa: The major recent work on Chiripa is described in Hastorf 1999. A summary is Stanish 2003:115-17.
- 230 Pukara: Stanish 2003:138-48, 156-60, 283-84. Stanish suggests that a drought that began in about 100 A.D. may have induced Pukara's collapse (157). But the drought may not have occurred—its existence has been deduced from a study of Lake Titicaca bottom sediments (Abbott et al. 1997). But the lake-sediment data, as the authors noted, conflict with previous ice-core studies; in addition, the depositional processes were sufficiently poorly understood that one could not judge when the putative dry periods began and ended.
- 230 Rise of Tiwanaku: Stanish 2003:chap. 8, 2001.
- 230 Tiwanaku as predatory state: Kolata 1993:81-86, 243-52 ("predatory," 243; "lower cost," 245).
- 231 Akapana: Interviews, Nicole Couture, Michael Moseley, Alexei Vranich; author's visit Cieza de León 1959:282 ("how human hands"); Kolata 1993:103-29. Wendell Bennet excavated at Akapana in the 1930s, but the first major excavations at Tiwanaku in general did not occur until the late 1960s, with the work of researchers from the Instituto Nacional de Arqueología de Bolivia, led by Carlos Ponce Sanginés. Ponce's work has come under fire, because he published little of his data and because he "restored" Tiwanaku landmarks inaccurately. In the 1980s Alan Kolata of the University of Chicago led a large team that produced the first comprehensive overview of the city and its environs.
- 232 Kalasasaya and Gateway of the Sun: Author's interviews, Couture, Vranich; Kolata 1993:143-49.
- 232 Isbell-Vranich vision of city: Isbell and Vranich 2004.
- 232 Lack of markets: Kolata 1993:172-76 ("a city was," "symbolically," 173).
- 233 Vranich picture of Tiwanaku: Interviews, Vranich; Vranich 2001; Vranich et al. 2001:150-52; Isbell and Vranich 2004.
- 235 Chimor history: Sakai 1998; Moseley and Cordy-Collins eds. 1990.